THE UNIVERSITY OF NEW SOUTH WALES

7b. Large Language Models

Never Stand Still | Faculty of Engineering | COMP9444 7b

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The Rise of Language Models

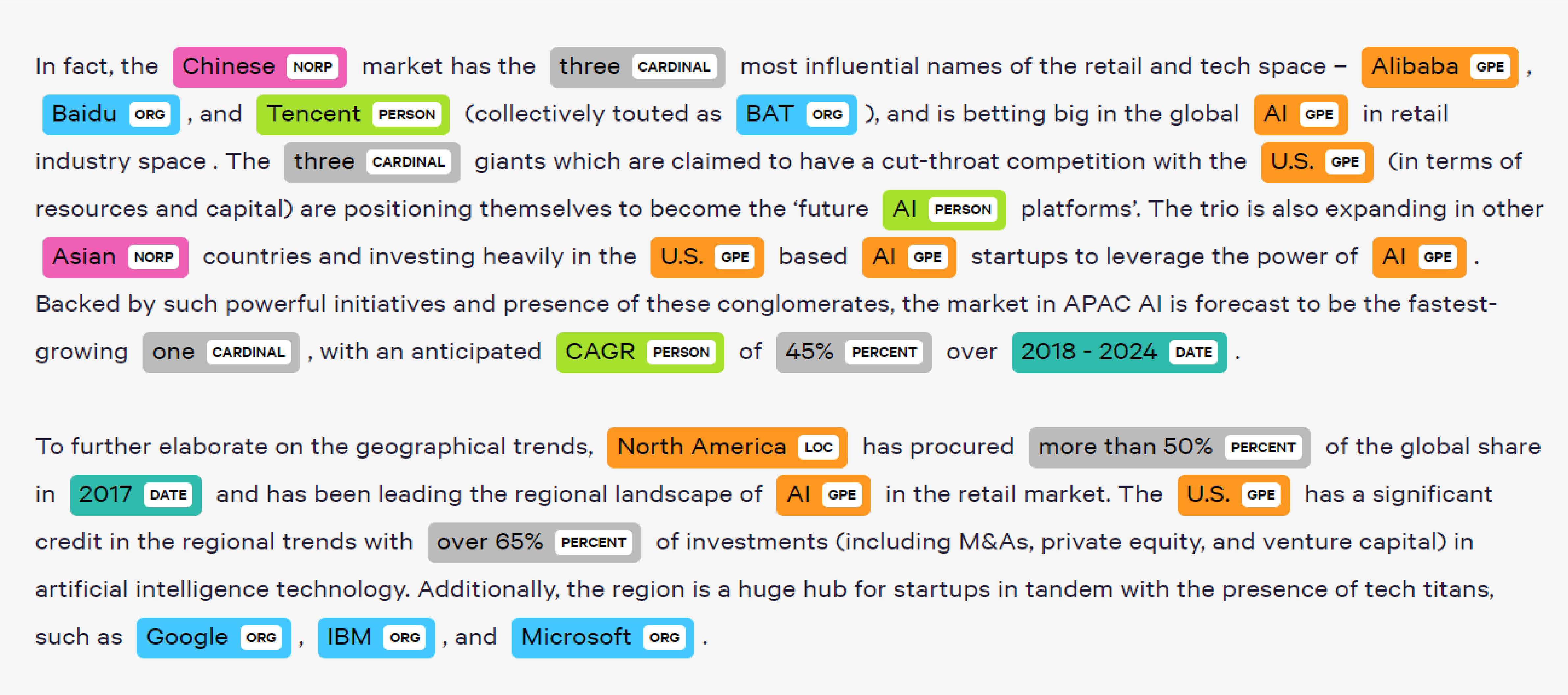
Photo Credit: Image by Free-photos from Pixabay. https://pixabay.com/photos/road-winding-street-bridge-1030789/

Motivation

- A robot wrote this entire article. Are you scared yet, human?
  https://www.theguardian.com/commentisfree/2020/sep/08/robot-wrote-this-article-gpt-3
- What It's Like To be a Computer: An Interview with GPT-3
  https://www.youtube.com/watch?v=PqbB07n_uQ4

Natural Language Processing

- Enabling machines to process, represent, understand, and generate languages



Source: Hugging Face. https://twitter.com/huggingface/status/1230870653194121216

NLP Applications

- Text Classification

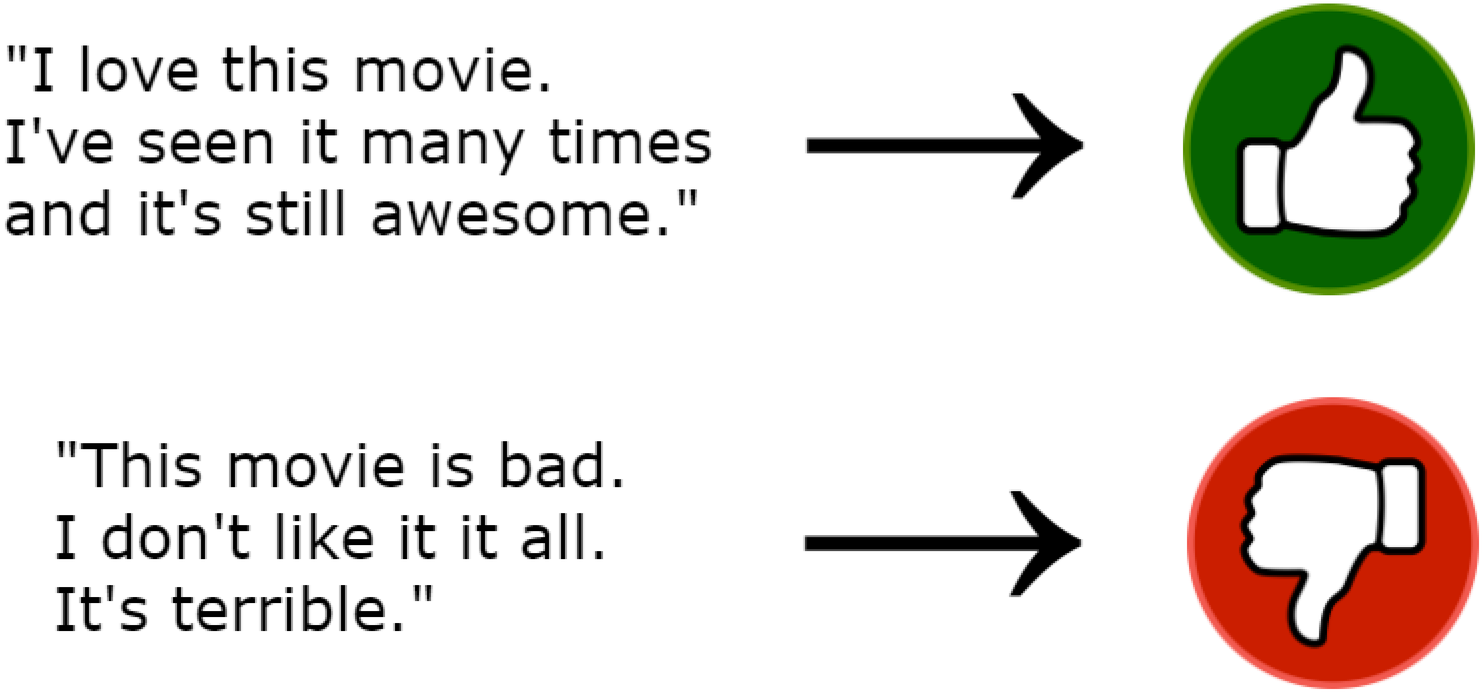
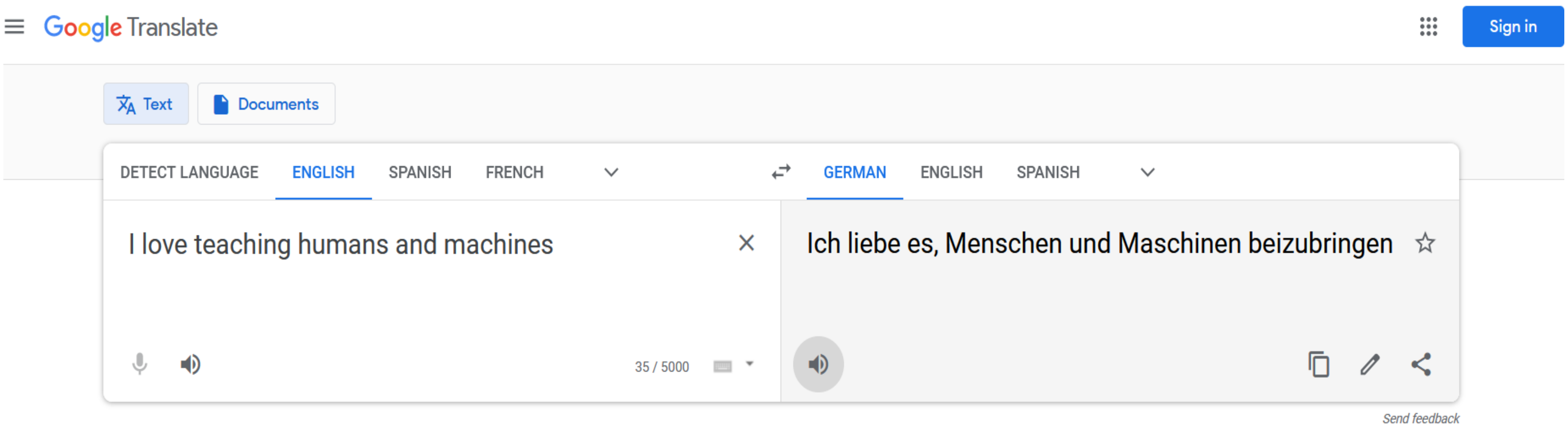


Image Credit: Sentiment Analysis. https://colab.research.google.com/github/shangeth/Google-ML-Academy/blob/master/2-Deep-Neural-Networks/2_9_ANN_Natural_Language_Processing.ipynb

NLP Applications

➢ Machine Translation



NLP Applications

- Question Answering/Comprehension

Passage Sentence

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under gravity.

Question

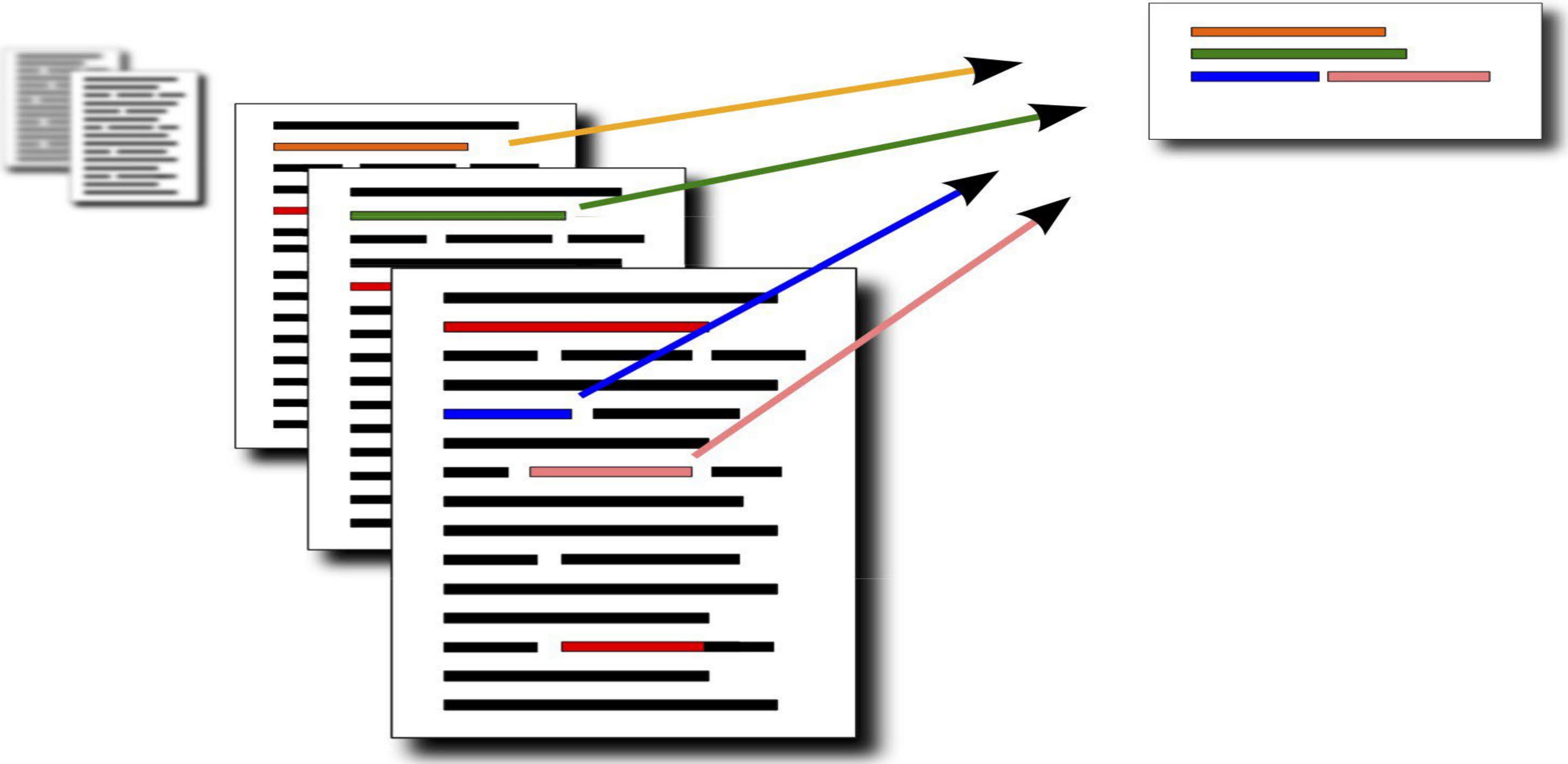
What causes precipitation to fall?

Answer Candidate

gravity

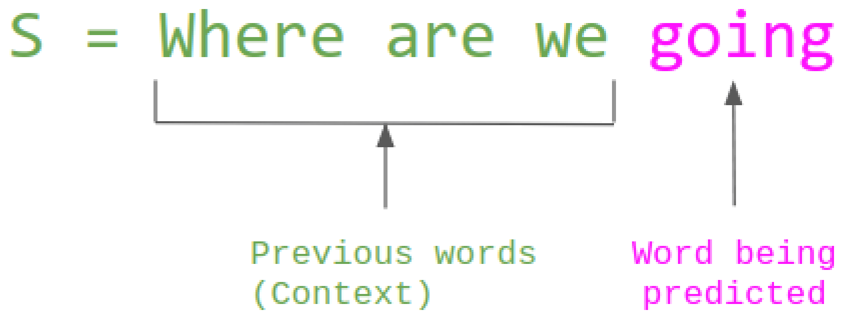
NLP Applications

- Text Summarization



Language Modelling

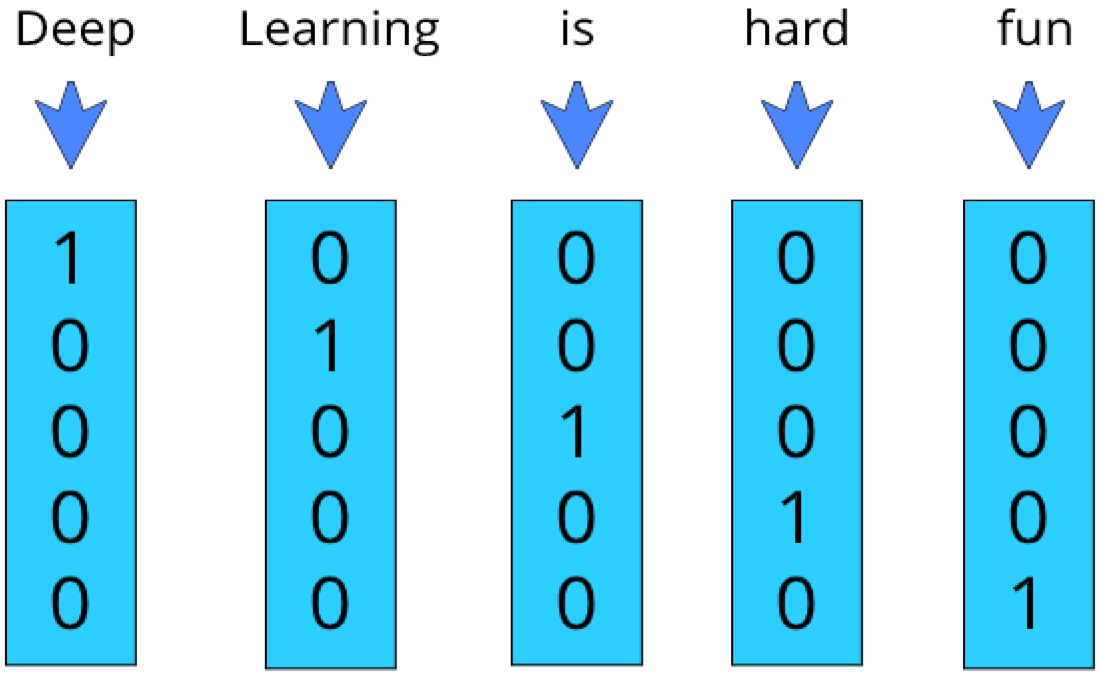
- A probability distribution over words or word sequences
- Probability distribution over strings of text
  - How likely is a string in a given “language”?



P(S) = P(Where) x P(are | Where) x P(we | Where are) x P(going | Where are we)

Representing text: One-hot Encoding

- One-hot vector of an ID is a vector filled with 0s, except for a 1 at the position associated with the ID.
  - For e.g., for vocabulary size D=10, the one-hot vector of word (w) having ID=4 is e(w) = [0 0 0 1 0 0 0 0 0 0]



- One-hot encoding makes no assumption abou word similarity
  - All words are equally similar/different from eac
- This is a natural representation to start with, though a poor one

Representing text: Count Vectorizer

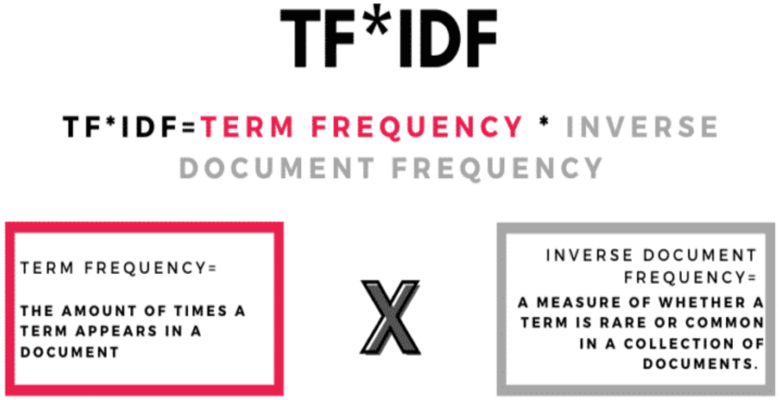
- Counts the number of occurrences of each word appearing in a document
- Converts a collection of text documents to a vector of term/token counts

	the	red	dog	cat	eats	food
1. the red dog	1	1	1	0	0	0
2. cat eats dog	0	0	1	1	1	0
3. dog eats food	0	0	1	0	1	1
4. red cat eats	0	1	0	1	1	0

Image Credit: Ishan Kotian. Fake news classification – NLP @Kaggle. https://www.kaggle.com/lykin22/fake-news-classification-nlp

Representing text: tf-idf vectorizer

- Stands for Term Frequency – Inverse Document Frequency (tf-idf)
- Common algorithm to transform text into a meaningful representation of numbers



$$\underbrace{w_{i,j}}_{\text{tf-idf score}} = \overbrace{tf_{i,j}}^{\text{\# occurrences of term in document}} \times \log \frac{\overbrace{N}^{\text{\# total documents}}}{\underbrace{df_j}_{\text{\# documents containing word}}}$$

- High TF: A term appears frequently in a document → might be important.
- Low IDF: A term appears in many documents → probably a common word (like "the", "is").
- High TF-IDF: A term is frequent in this document but rare in the overall corpus → very informative.

Representing text as vectors

- One-hot encoding, count and tf-idf vectors are **sparse** and **long**
  - Most values are zeros
  - Length of vector is equal to size of vocabulary (can be more than 50,000)

- Alternative is to look for **dense vectors.** These are
  - short and dense
  - easier to use a features in machine learning (less weights to tune)
  - can capture context

Word Embeddings

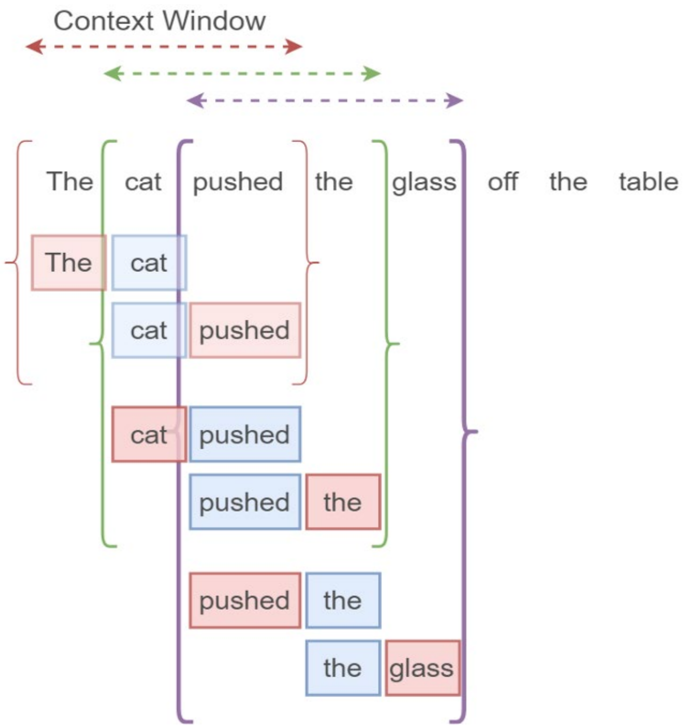
- Instead of counting how often each word w occurs near “university”, train a classifier on a binary prediction task:
  - Is w likely to show up near “university”?

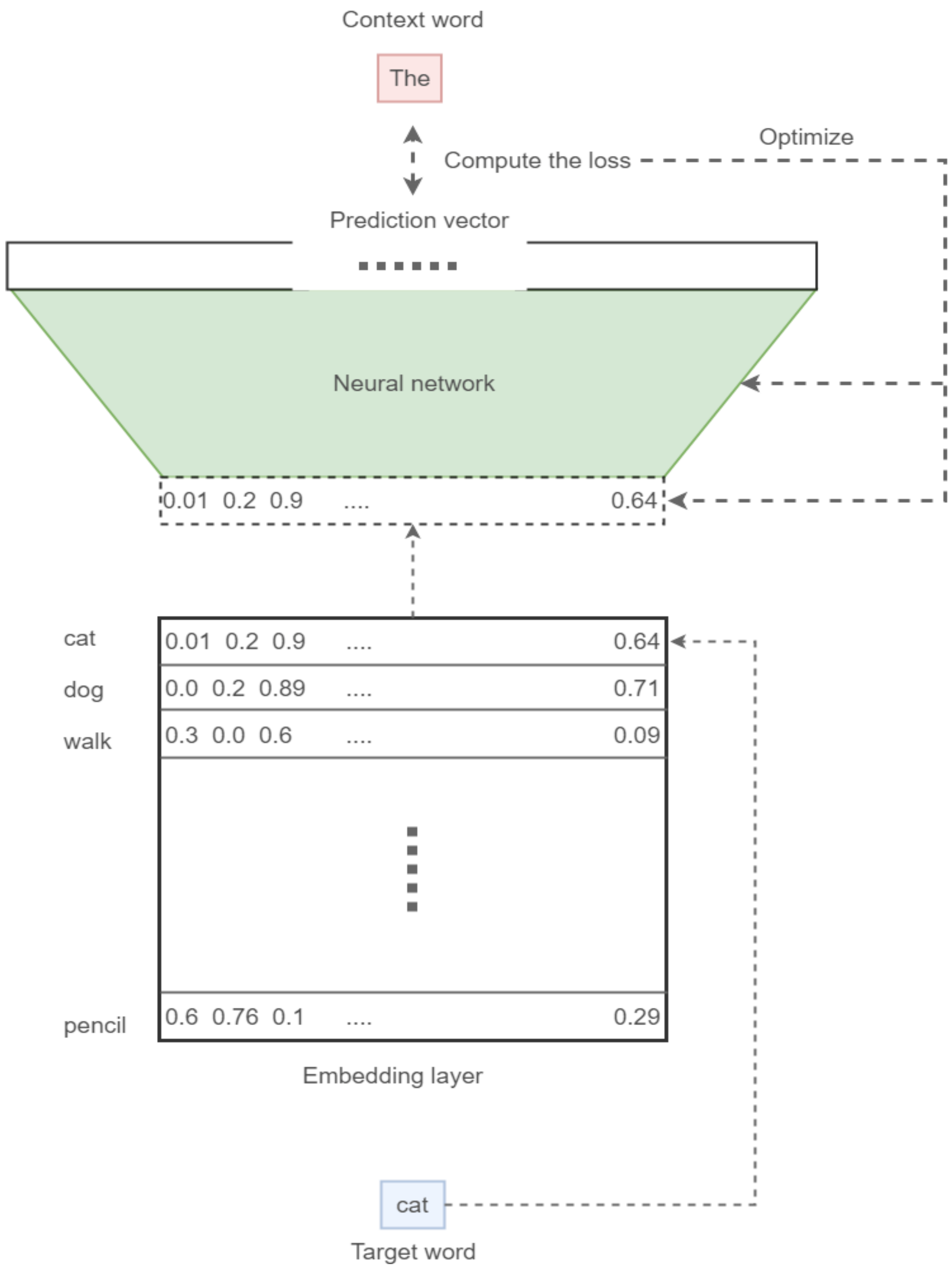
- We don’t actually care about this task, but we take the learned classifier weights as the word embeddings
- Uses free-form text as implicitly supervised training data
  - A word s near “university” acts as gold standard “ground-truth” to the question
  - No need for hand-labeled supervision

- Word vector algorithms use the context of the words to learn numerical representations for words, so that words used in the same context have similar looking word vectors.

Word2Vec

- “You should know a word by the company it keeps”
  – J.R. Firth
- Uses CBOW or Skip-gram model to train the network
- Word2Vec uses skip-gram neural network to predict neighbor context

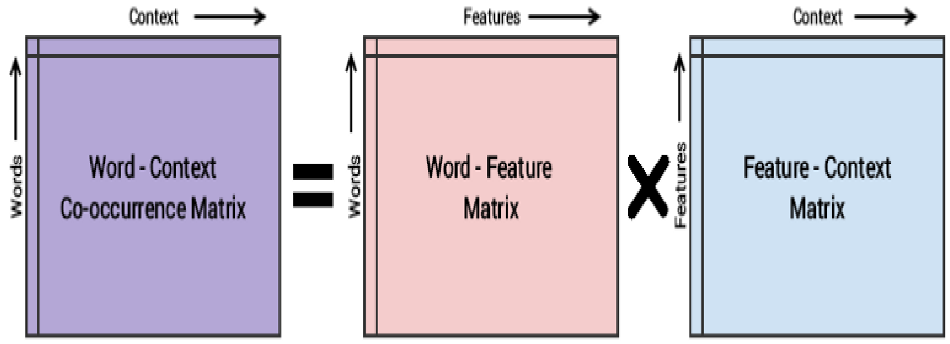


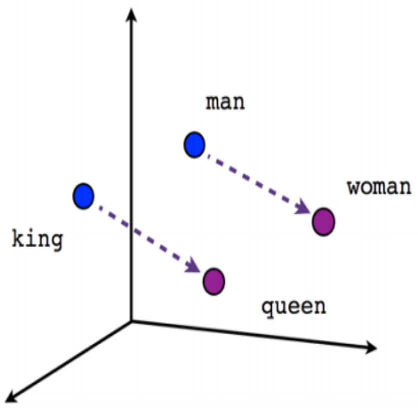


Credit: Thushan Ganegedara. Intuitive guide to understanding Word2Vec. https://towardsdatascience.com/light-on-math-machine-learning-intuitive-guide-to-understanding-word2vec-e0128a460f0f

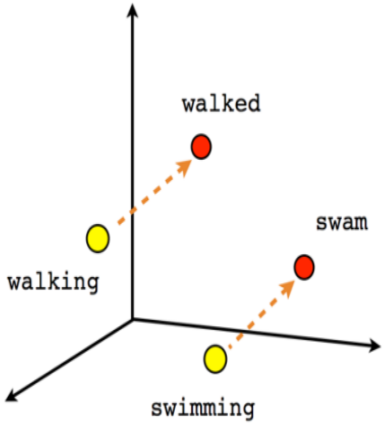
GloVe

- Word2Vec rely on local statistics (local context information of words)
- GloVe captures both local and global statistics (word co-occurrence) to obtain word vectors
- Can derive semantic relationships between words from the co-occurrence matrix
- Requires upfront pass-through entire dataset

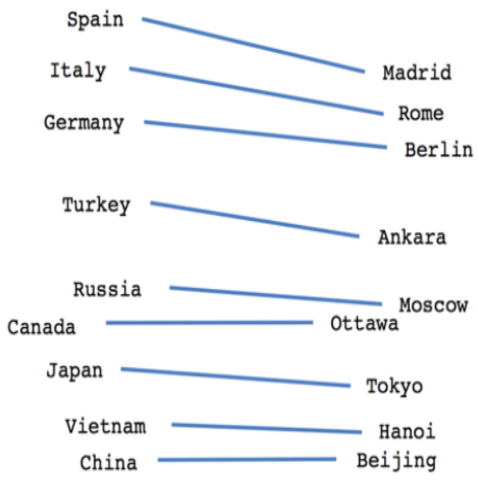




Male-Female



Verb tense



Country-Capital

Statistical Language Modeling

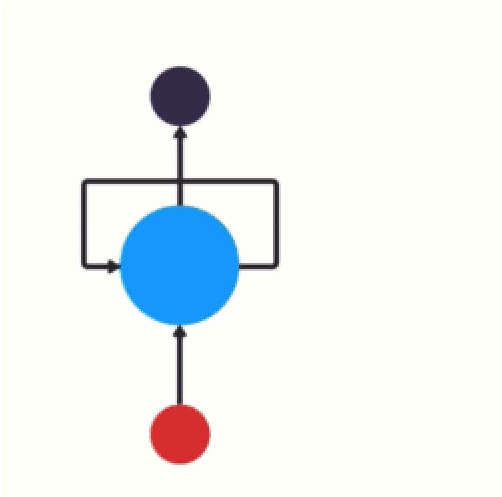
- Models that assign probabilities to the sequences of words
- Focus on n-gram
  - 2-gram (bigram). E.g., “going to”, “on the”
  - 3-gram (trigram). E.g., “what are you”, “please turn your”

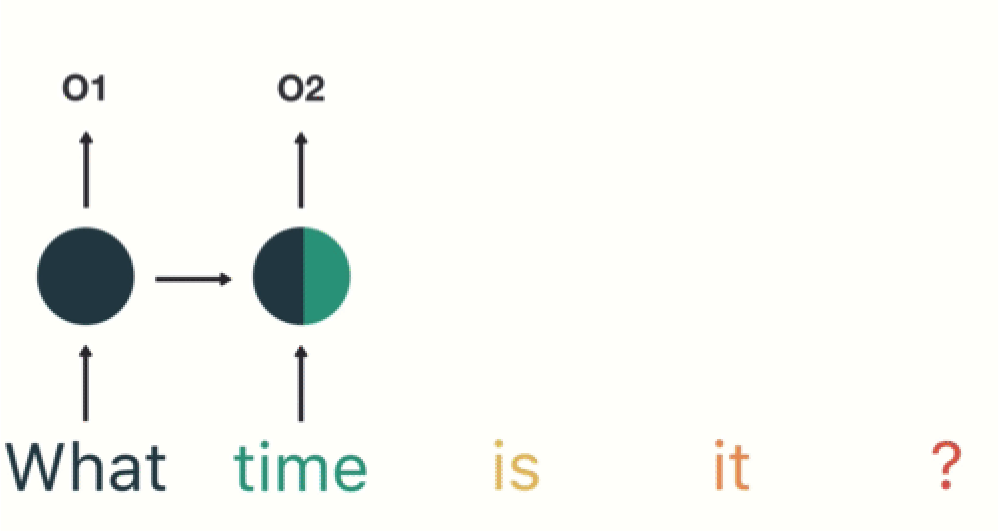
- Naïve approach:
  - Calculate the probability of word w, given history, h: P(w/h)
  - A computation challenge with significant size of corpus
  - Do approximation using chain rule

- Bigram language model
  - Approximates the probability of a word given all the previous words by using the conditional probability of one preceding word
  - Based on Markov assumption: we can predict the probability of some future unit without looking too far in the past
  - Use Maximum Likelihood Estimation (MLE) to estimate the probability function

Recurrent Neural Networks (RNNs)

- A class of artificial neural networks suitable for modeling sequential or time-series data
- Exhibit dynamic behavior where connections between nodes form a directed graph along a temporal sequence
- RNNs have a looping mechanism that acts as a highway to allow information to flow from one step to the next. This information is the hidden state, which is a representation of previous inputs.

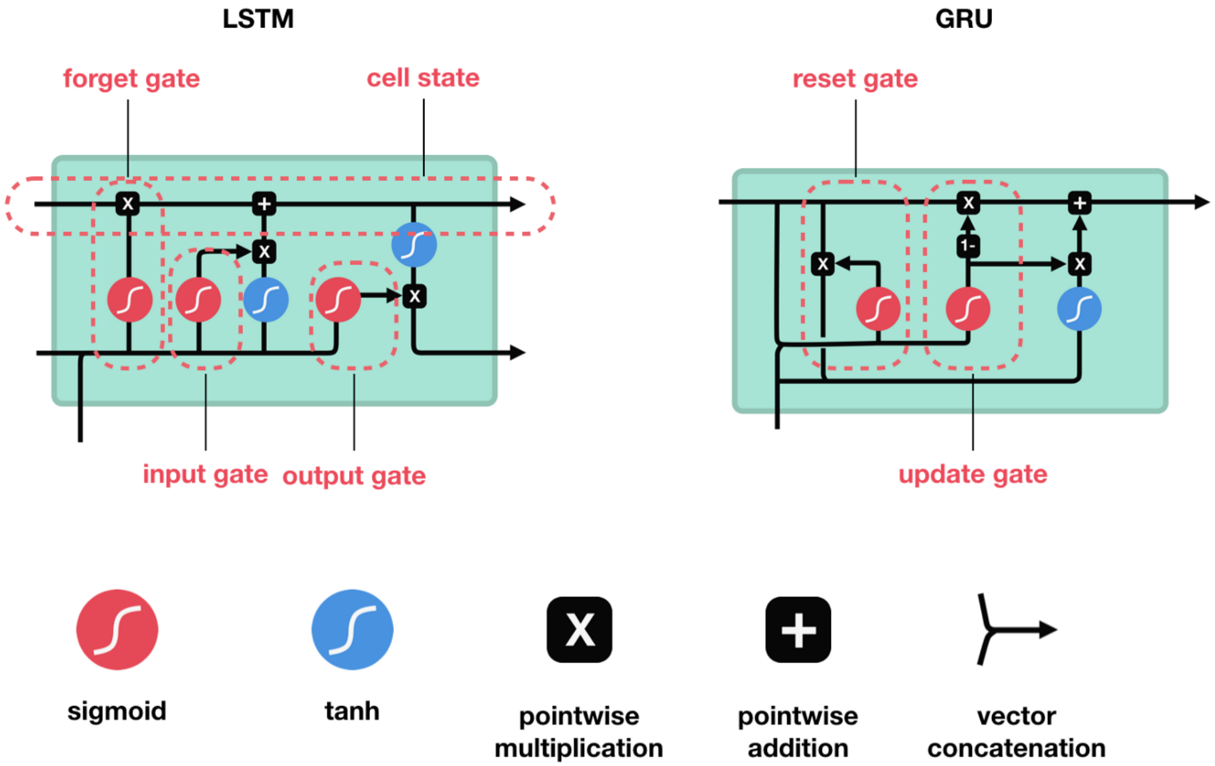




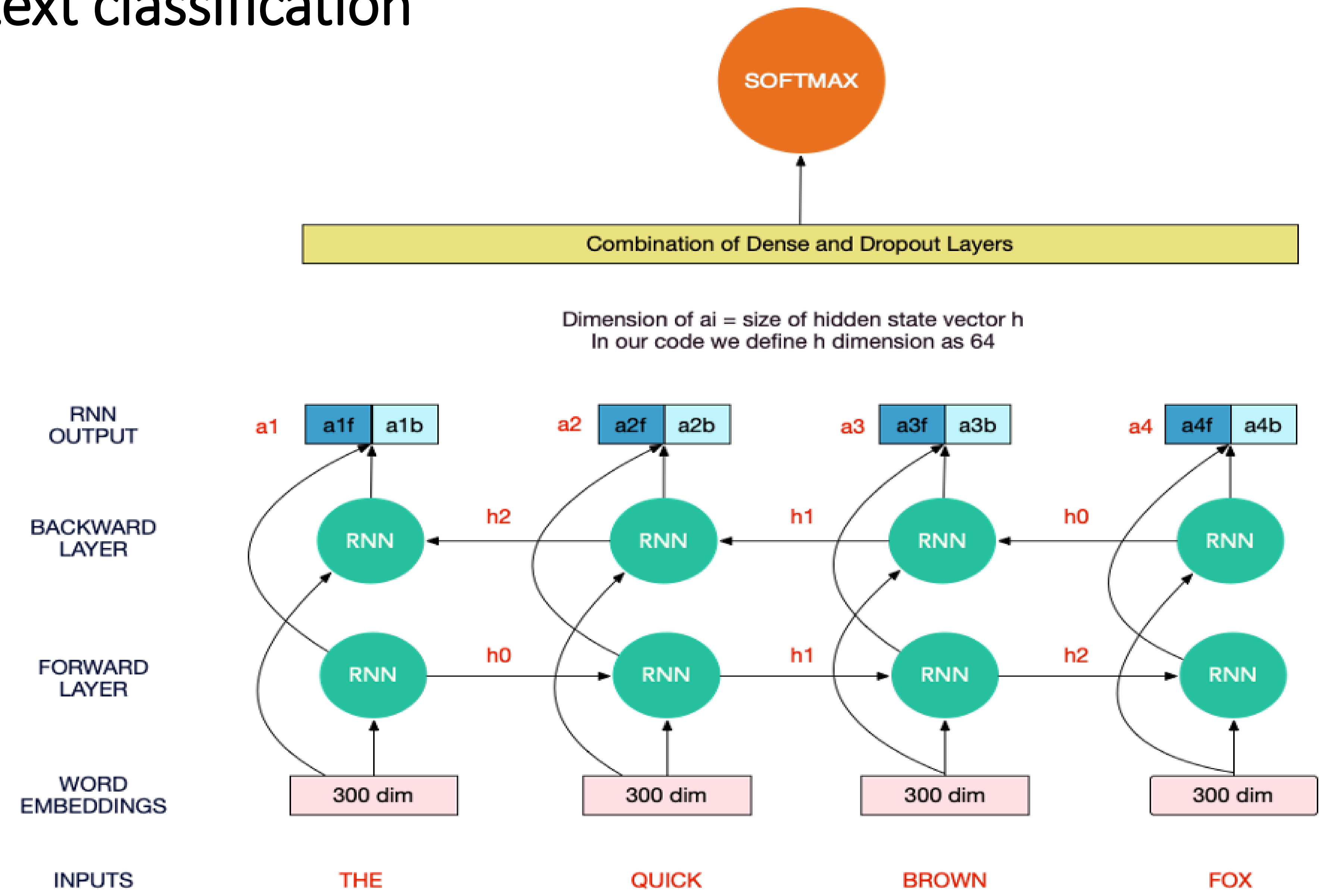
Credit: Michael Phi. Illustrated guide to RNNs. https://towardsdatascience.com/illustrated-guide-to-recurrent-neural-networks-79e5eb8049c9

LSTMs/GRUs

- Vanilla RNNs suffers from vanishing gradient problem
  - As the RNNs processes more steps, it has troubles retaining information from previous steps.
  - Due to back-propagation, the earlier layers fail to do any learning as the internal weights are barely being adjusted due to extremely small gradients.
  - Does not learn the long-range dependencies across time steps
- LSTMs and GRUs are two special RNNs, capable of learning long-term dependencies using mechanisms called **gates**.
- These gates are different tensor operations that can learn what information to add or remove to the hidden state.

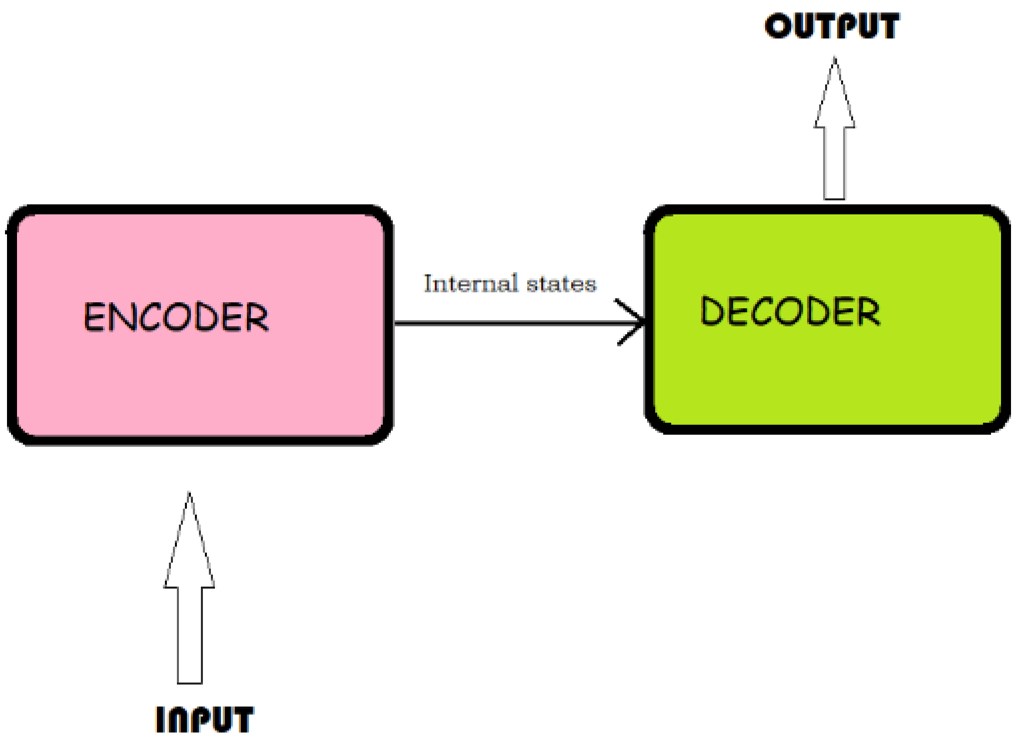


End-to-End text classification



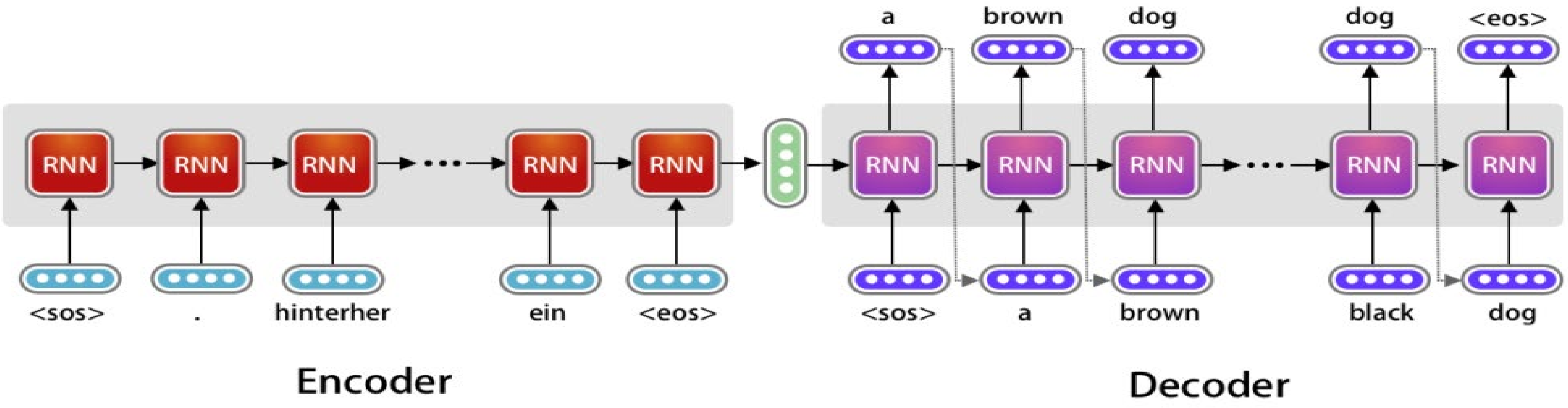
Seq2Seq model

- Model has two parts: Encoder and Decoder (Encoder-Decoder Framework)



Seq2Seq used for various NLP applications

- Machine Translation



- Automatic Email Reply

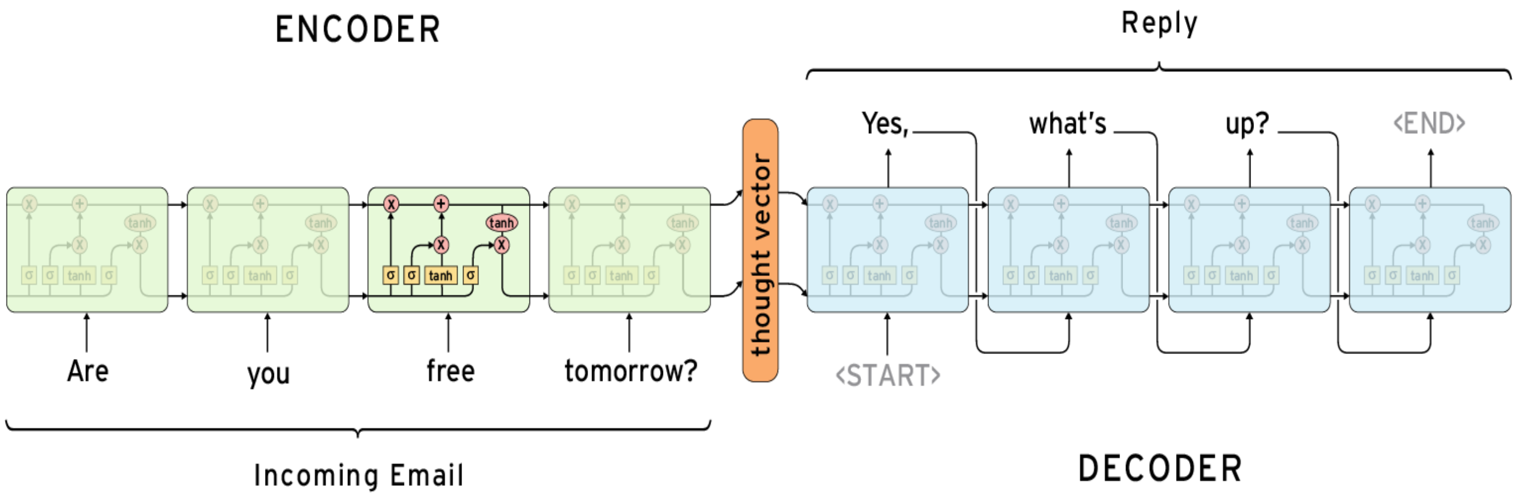
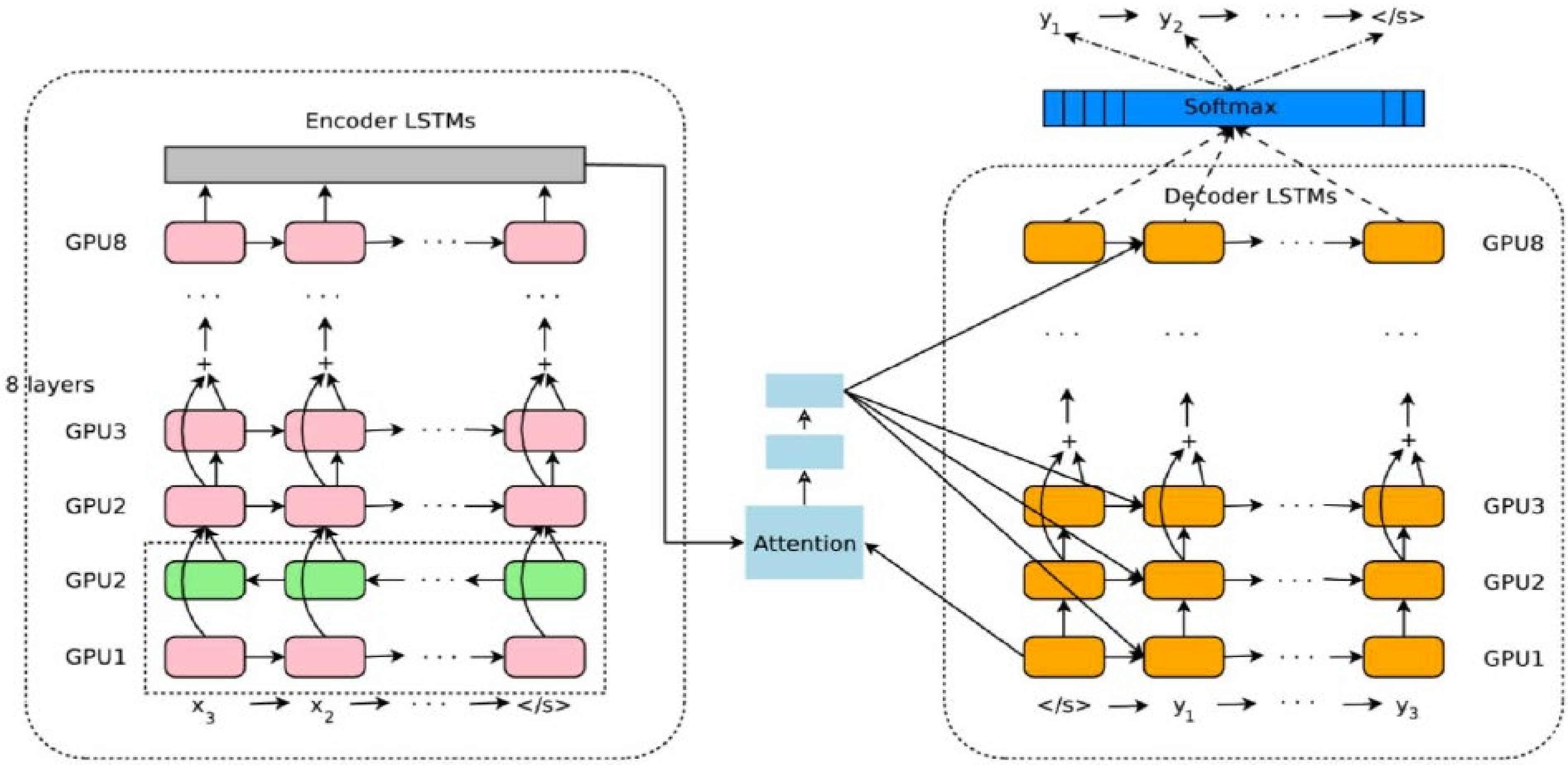


Image Credit: Machine Translation using Recurrent Neural Network and PyTorch. http://www.adeveloperdiary.com/data-science/deep-learning/nlp/machine-translation-recurrent-neural-network-pytorch/

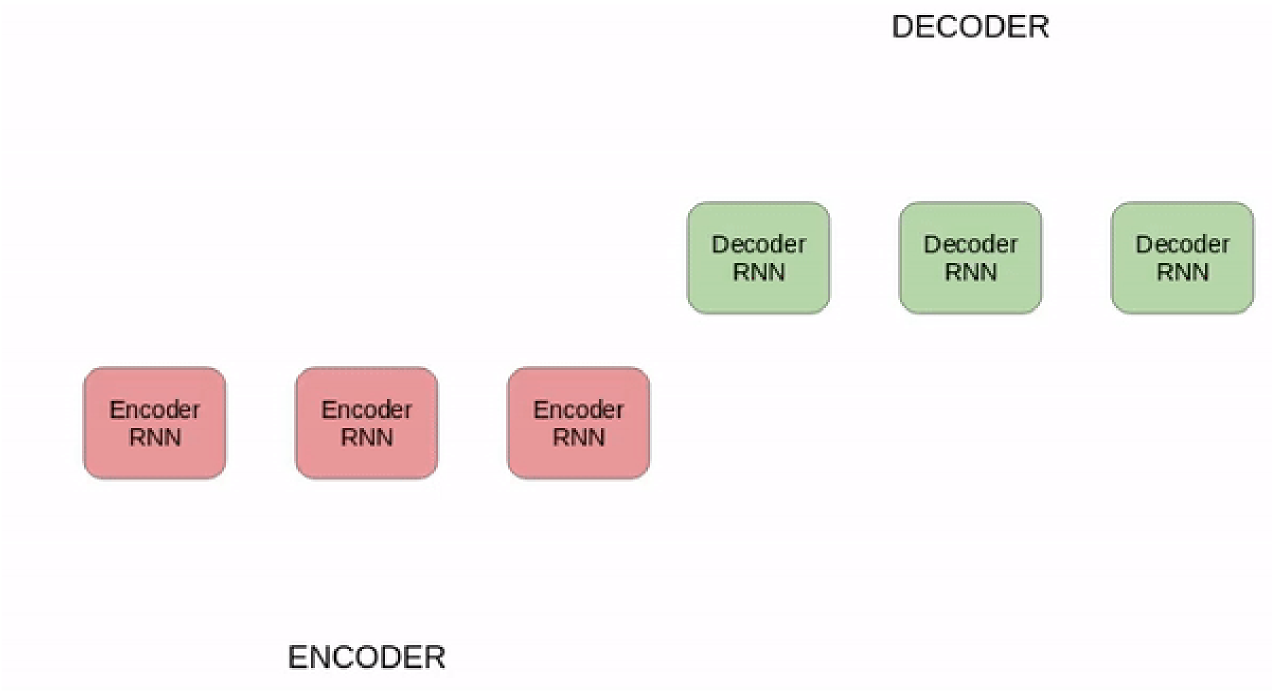
Seq2Seq used for various NLP applications

- Google's Multilingual Neural Machine Translation



Issues with RNNs for seq2seq tasks

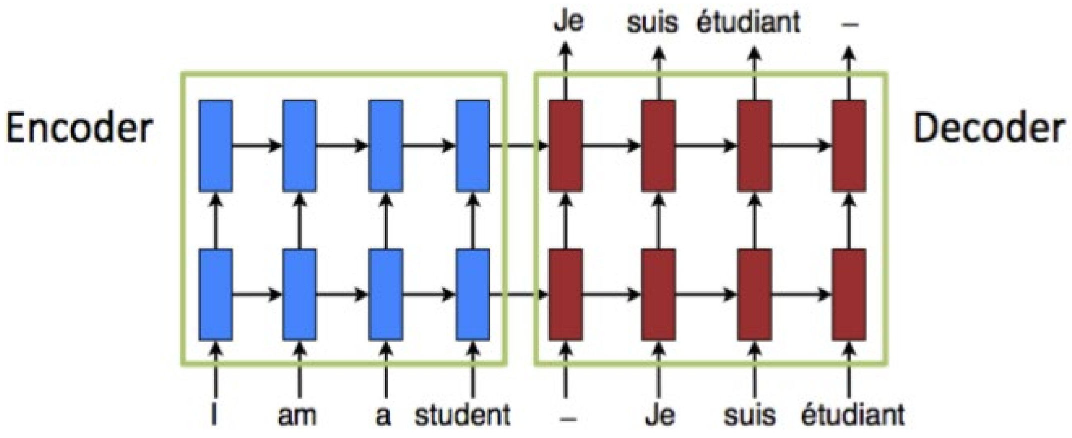
- Despite being very successful for various NLP tasks, there were challenges:
  - dealing with long-range dependencies
  - the sequential nature of the architecture prevents parallelization
- Attention mechanism helped to overcome first issue to certain extent



Credit: Encoder-Decoder Architecture for Machine Translation. https://www.analyticsvidhya.com/blog/2019/06/understanding-transformers-nlp-state-of-the-art-models/

Neural Machine Translation by jointly Learning to Align and Translate

- NMT attempts to build and train a single, large neural network that reads a sentence and outputs a correct translation.
- The NMT system is a seq2seq model (Encoder-Decoder framework)
- Issue: The encoder-decoder framework compresses all the necessary information of a source sentence into a fixed-length vector.



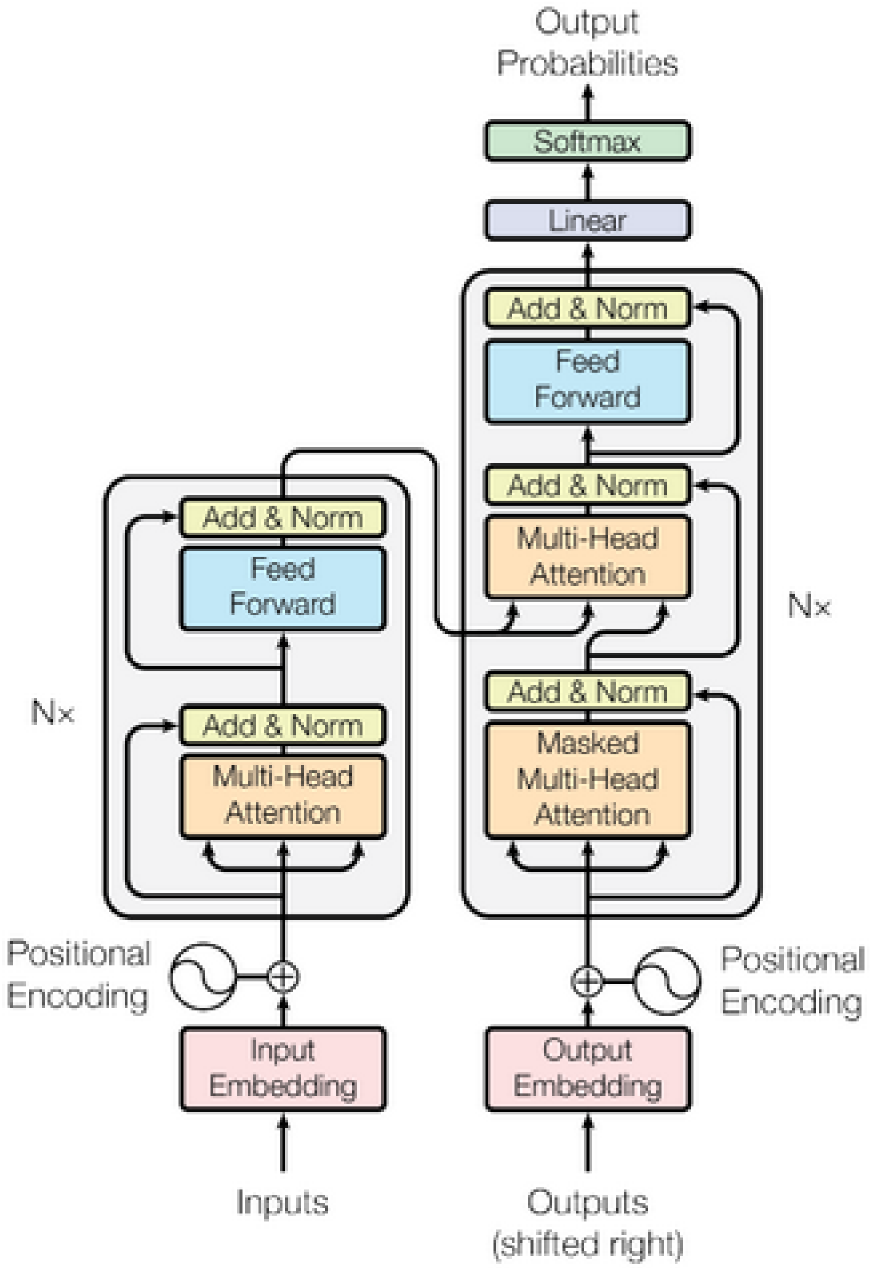
- Cho et al. (2014) showed that the performance of a basic encoder-decoder deteriorates as the length of an input sentence increases.
- Bahdanau et al. (2015) proposed an extension to NMT which learns to align and translate jointly.
- The basic idea of “attention” is that at each time the model generates a word in a translation, it searches for a set of positions in a source sentence where the most relevant information is concentrated
- The model predicts a target word based on the context vectors associated with these source positions and all the previous generated target words

Credit: Bahdanau et al. (2015). Neural Machine Translation by jointly learning to Align and Translate. ICLR 2015.

Transformers

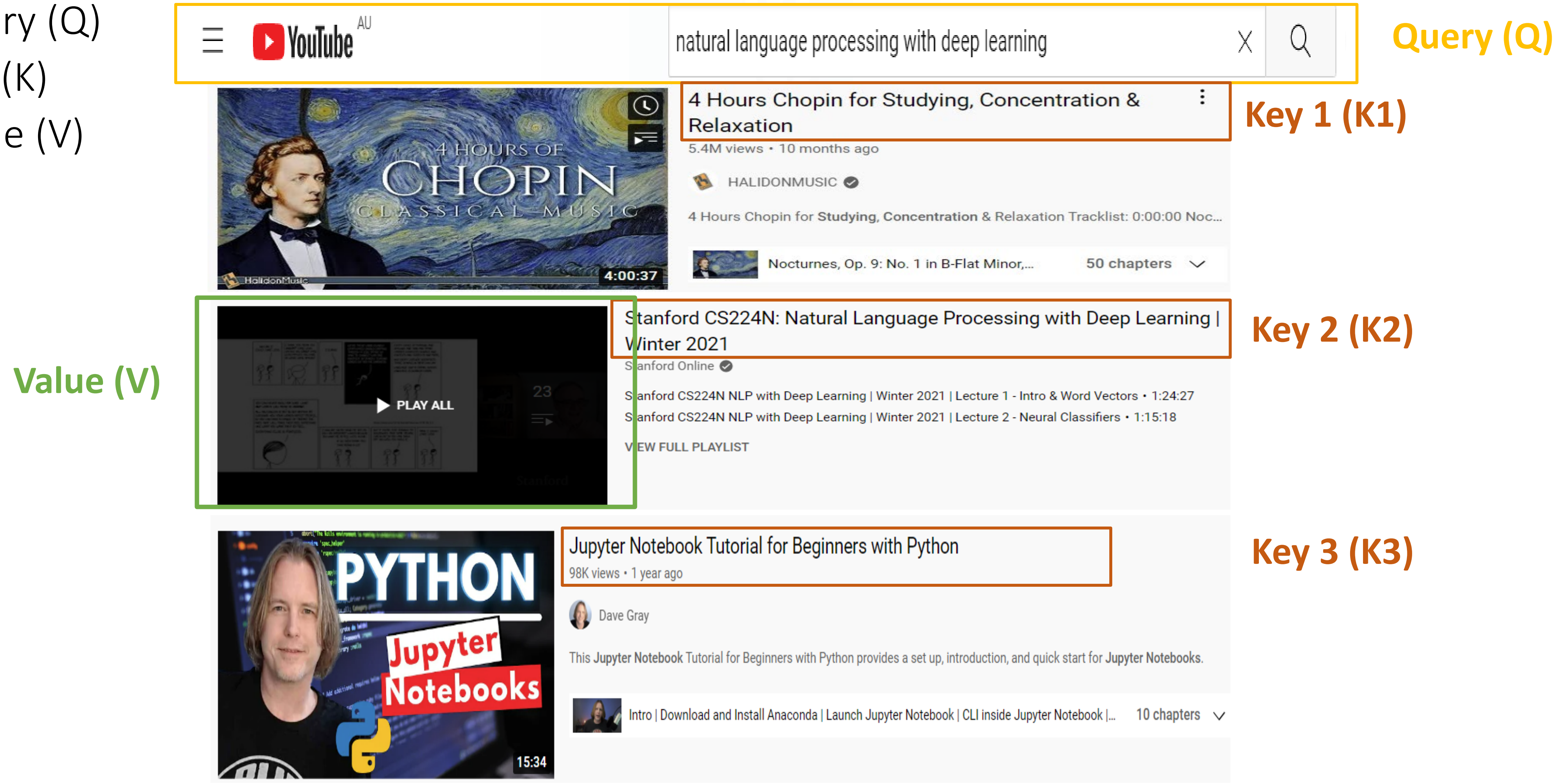
- Attention is All You Need
- Novel architecture relies entirely on **self-attention** to compute representations of its input and output without using sequential RNNs or convolutions.
- Aim is to solve seq2seq tasks while handling long-range dependencies

"Griezmann's announcement comes as a bit of a shock. After enduring the drama surrounding his potential last summer, many thought he was committed to Atletico for more than a year, but the Frenchman seems to have changed his mind."



Self-attention in transformers

- Understanding self-attention with Search
  - Query (Q)
  - Key (K)
  - Value (V)



Self-attention in transformers

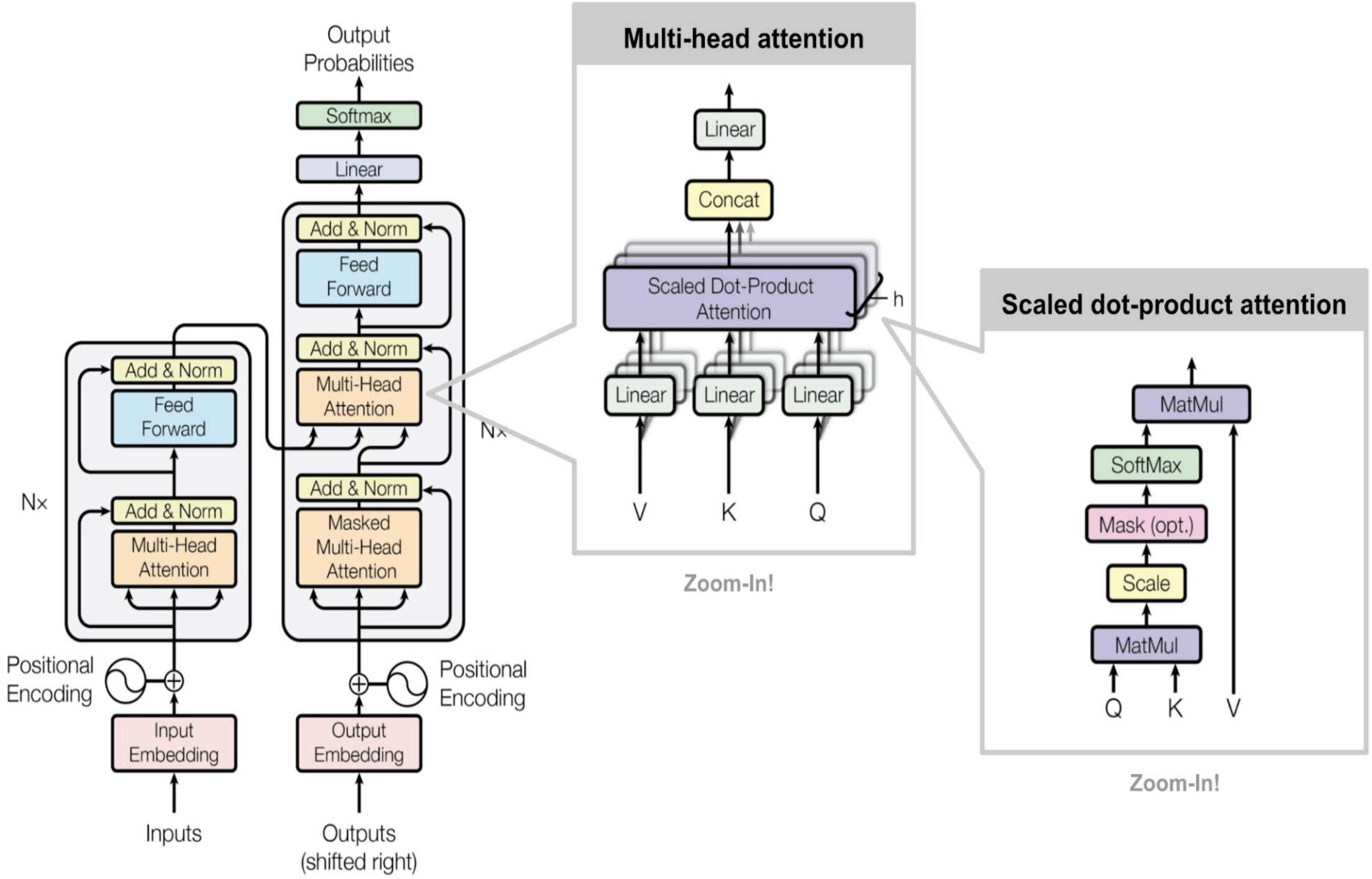
Self-attention:

Step 1: Encode position information

Step 2: Extract query (Q), key (K), and value (V

Step 3: Compute attention weighting

Step 4: Extract features with high attention



BERT

- **B**idirectional **E**ncoder **R**epresentations from **T**ransformers
- Idea to have **deeply bidirectional, unsupervised language representation**
- Example sentence: "I accessed the bank account"
  - Unidirectional contextual model would represent "bank" based on "I accessed the"
  - BERT represent "bank" based on both its previous and next context – "I accessed the --- account"
- Why bidirectional in BERT possible?
  - Not possible to train bidirectional models by simply conditioning each word on its previous and next words, since this would allow the word that's being predicted to indirectly "see itself" in a multi-layer model.
  - BERT rely on two tasks
    - Masking words in the input
    - Next sentence prediction

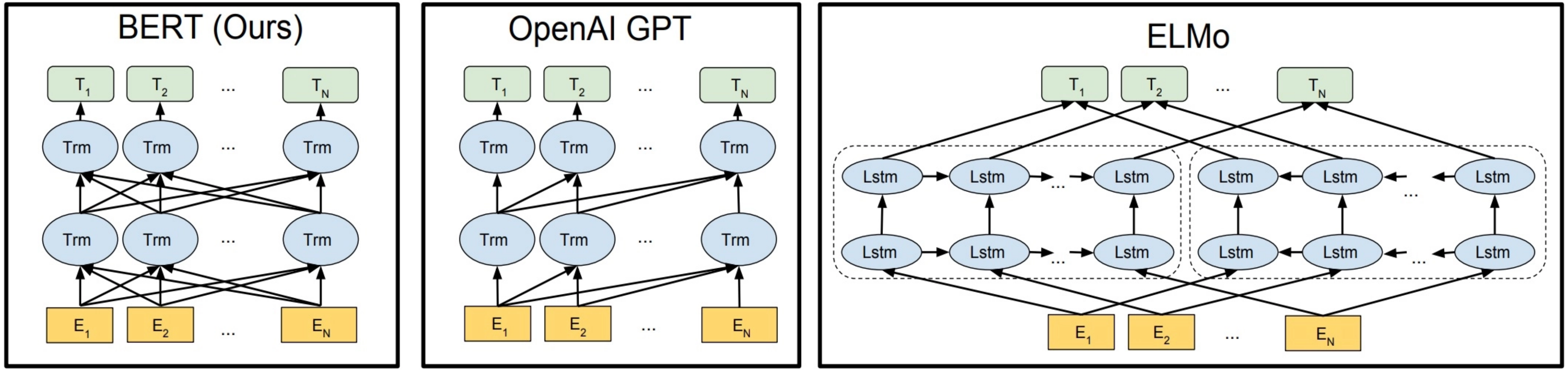
Input: The man went to the [MASK]$_1$. He bought a [MASK]$_2$ of milk .
Labels: [MASK]$_1$ = store; [MASK]$_2$ = gallon

Sentence A = The man went to the store.
Sentence B = He bought a gallon of milk.
Label = IsNextSentence

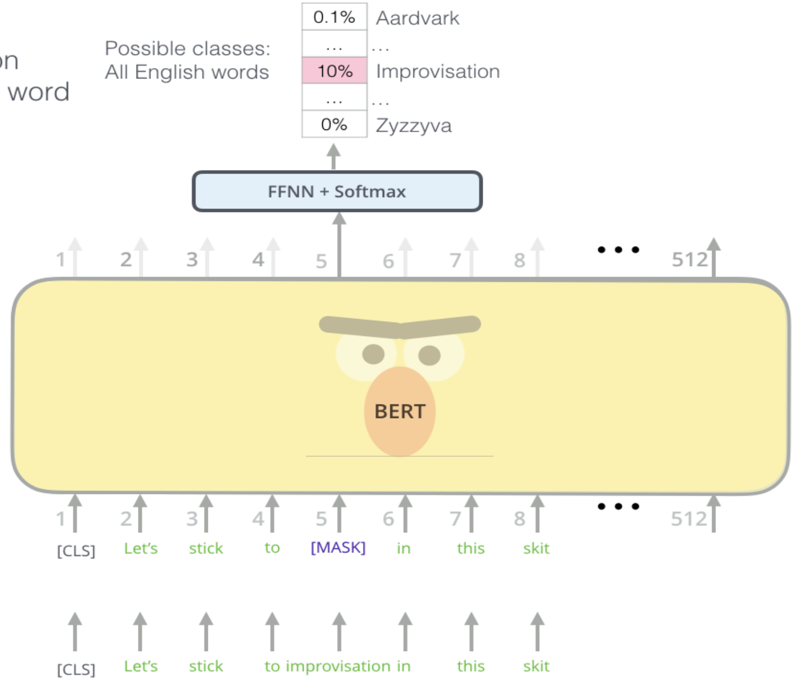
Sentence A = The man went to the store.
Sentence B = Penguins are flightless.
Label = NotNextSentence

Credit: Devlin et al., BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.
Google AI Blog. https://ai.googleblog.com/2018/11/open-sourcing-bert-state-of-art-pre.html

BERT



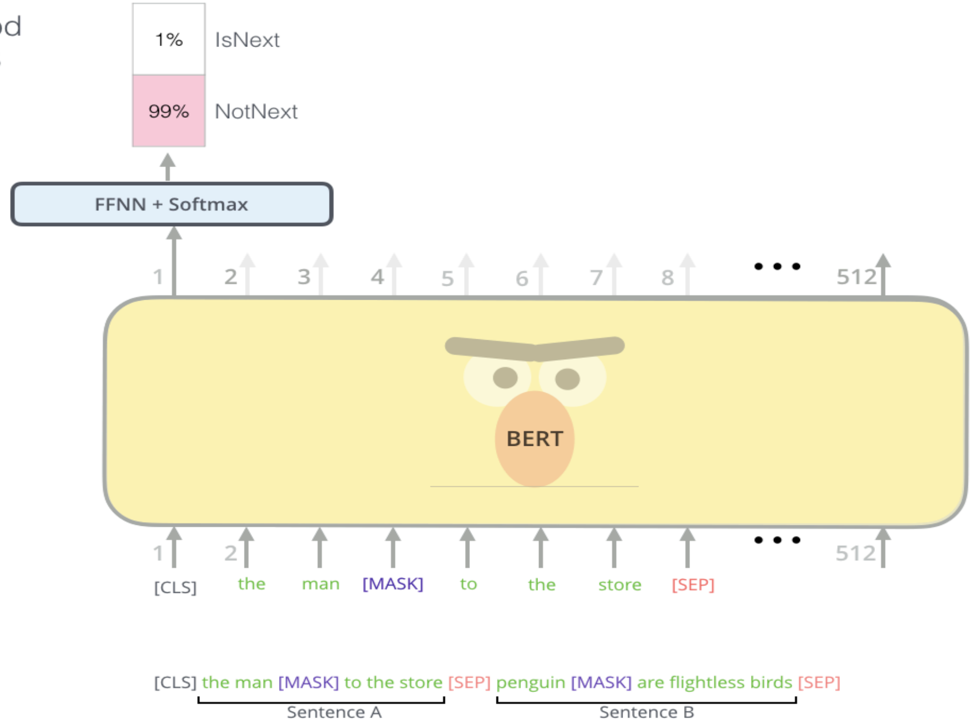
Use the output of the masked word's position to predict the masked word



Randomly mask 15% of tokens

Input

Predict likelihood that sentence B belongs after sentence A



Tokenized Input

Input

What is Foundation Model (1)

NVIDIA:

Foundation models are AI neural networks trained on **massive unlabeled datasets** to handle a **wide variety** of jobs from translating text to analyzing medical images.

AWS:

Trained on **massive datasets**, FMs are large deep learning neural networks that have changed the way data scientists approach machine learning. Rather than develop AI from scratch, data scientists use a foundation model as a starting point to develop ML models that power **new applications more quickly** and cost-effectively.

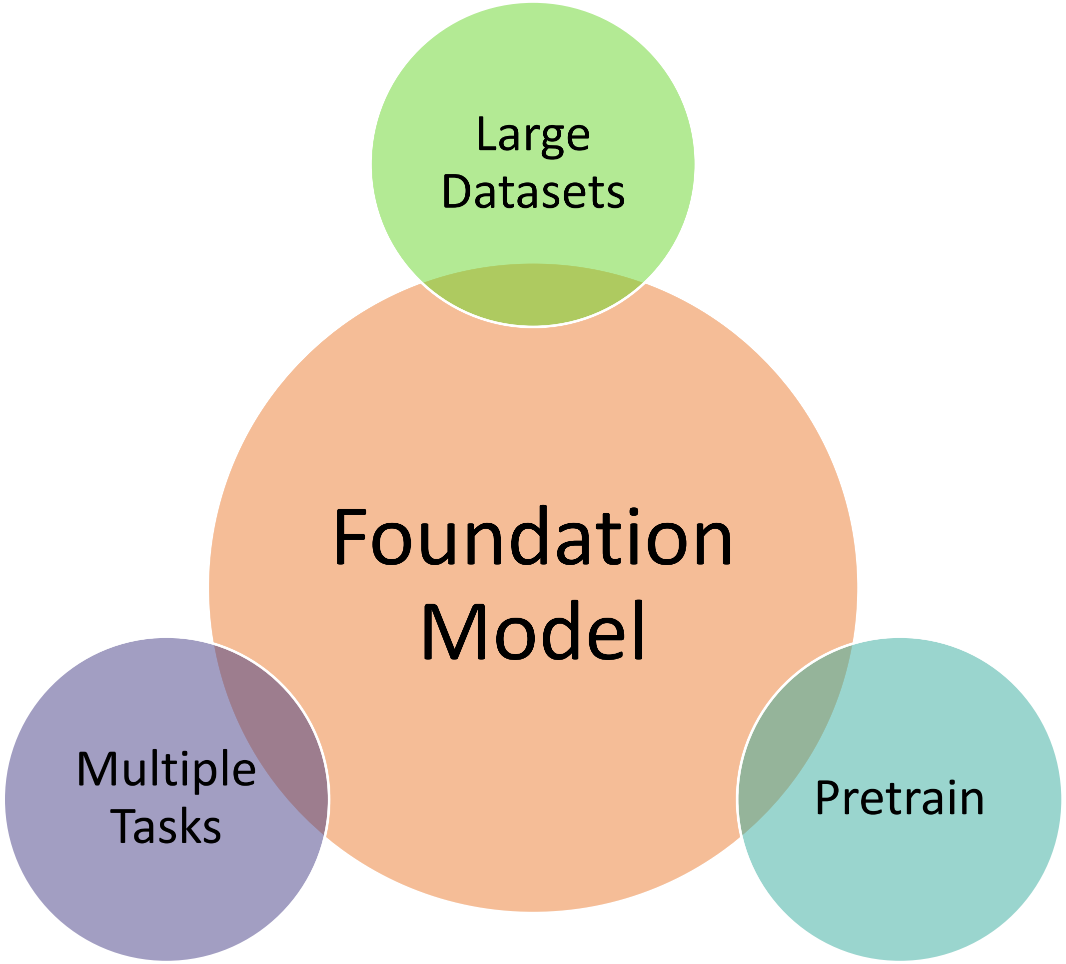
What is Foundation Model (2)

Center for
Research on
Foundation
Models

Train one model on a huge amount of data and adapt it to many applications. We call such a model a foundation model.

https://crfm.stanford.edu/

What is Foundation Model (3)



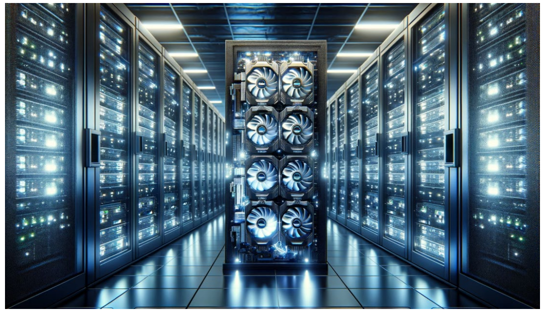
Large Models:

Large Datasets;
Large-Scale Parameters;
Large-Scale Downstream Tasks

How Large?

The internet scale, 10TB of text

+



6,000 GPUs for 12 days, $2M USD

pytorch_model-00001-of-00015.bin	pickle	9.85 GB	LFS
pytorch_model-00002-of-00015.bin	pickle	9.8 GB	LFS
pytorch_model-00003-of-00015.bin	pickle	9.97 GB	LFS
pytorch_model-00004-of-00015.bin	pickle	9.8 GB	LFS
pytorch_model-00005-of-00015.bin	pickle	9.8 GB	LFS
pytorch_model-00006-of-00015.bin	pickle	9.8 GB	LFS
pytorch_model-00007-of-00015.bin	pickle	9.97 GB	LFS
pytorch_model-00008-of-00015.bin	pickle	9.8 GB	LFS
pytorch_model-00009-of-00015.bin	pickle	9.8 GB	LFS
pytorch_model-00010-of-00015.bin	pickle	9.8 GB	LFS
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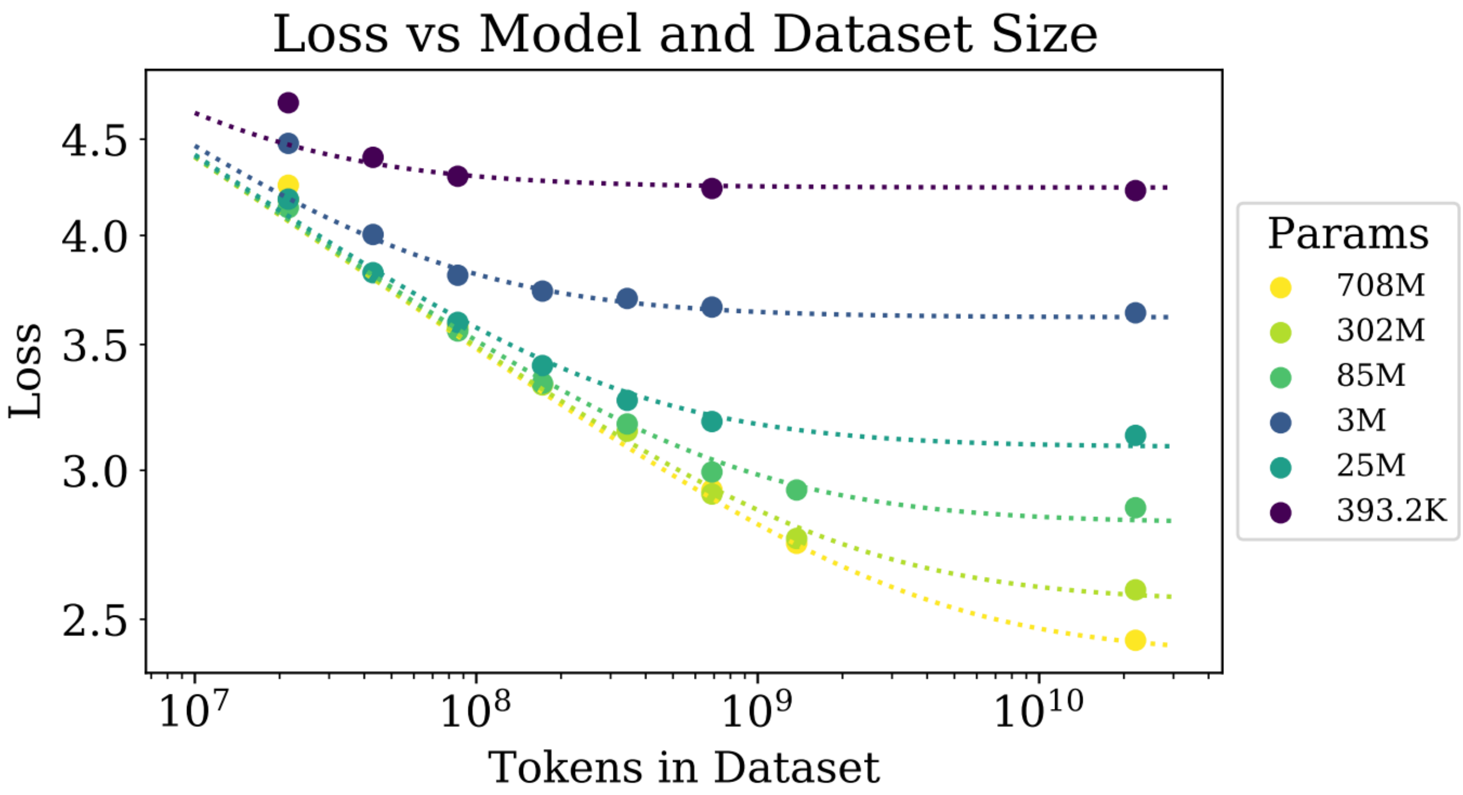
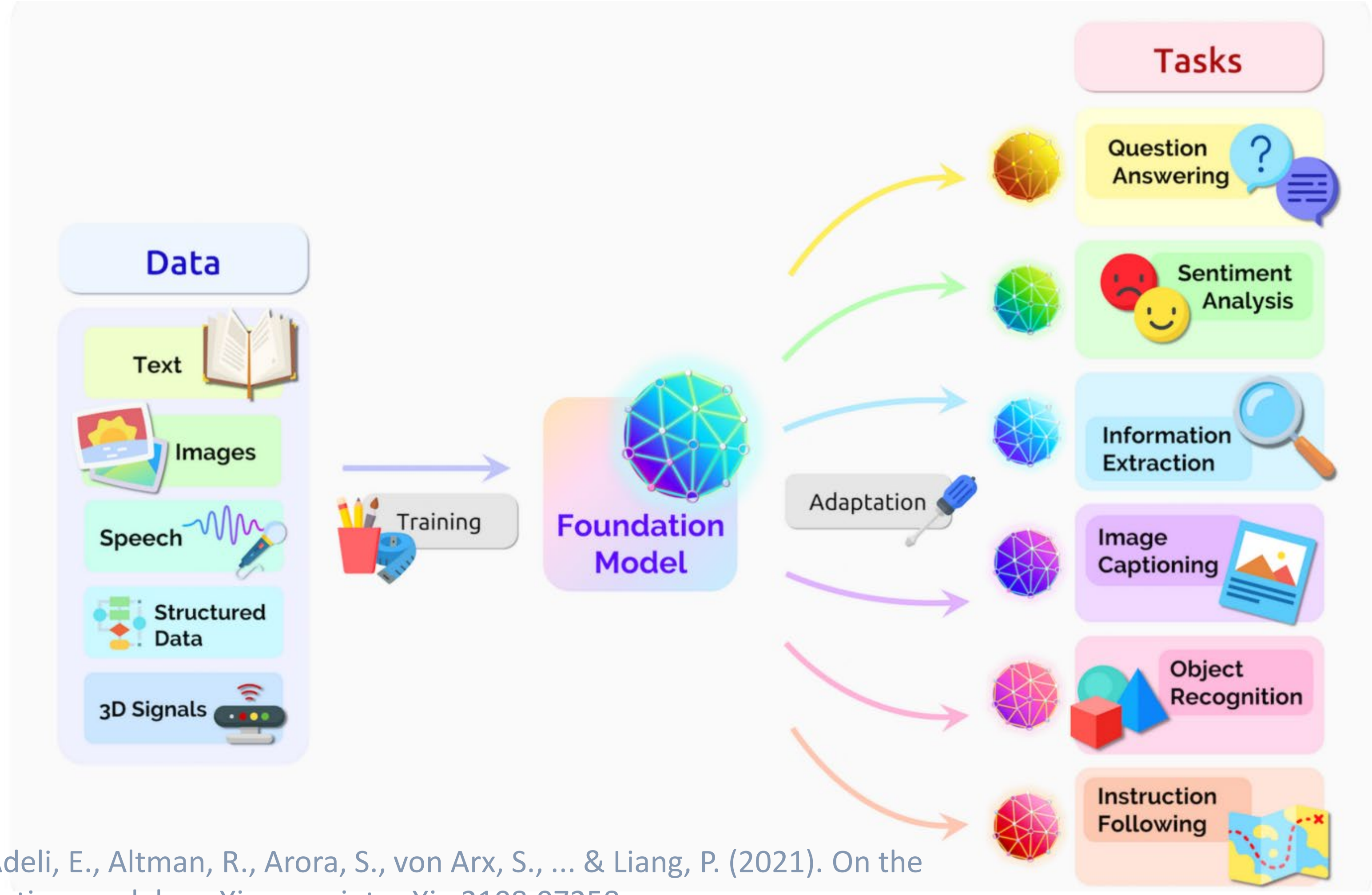
pytorch_model-00013-of-00015.bin	pickle	9.8 GB	LFS
pytorch_model-00014-of-00015.bin	pickle	9.5 GB	LFS
pytorch_model-00015-of-00015.bin	pickle	524 MB	LFS

140GB File (LLAMA2 70B Parameters)


Loss vs Model and Dataset Size
Loss
4.5
4.0
3.5
3.0
2.5
10^7
10^8
10^9
10^{10}
Tokens in Dataset
Params
708M
302M
85M
3M
25M
393.2K

What is Foundation Model (4)

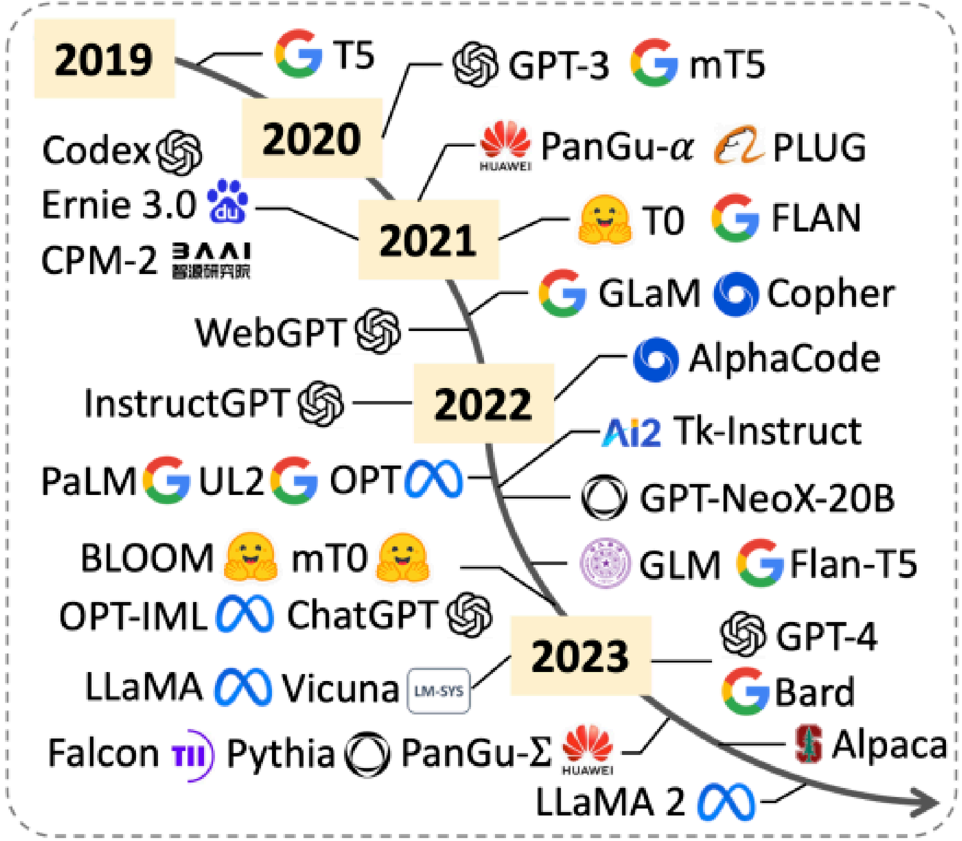


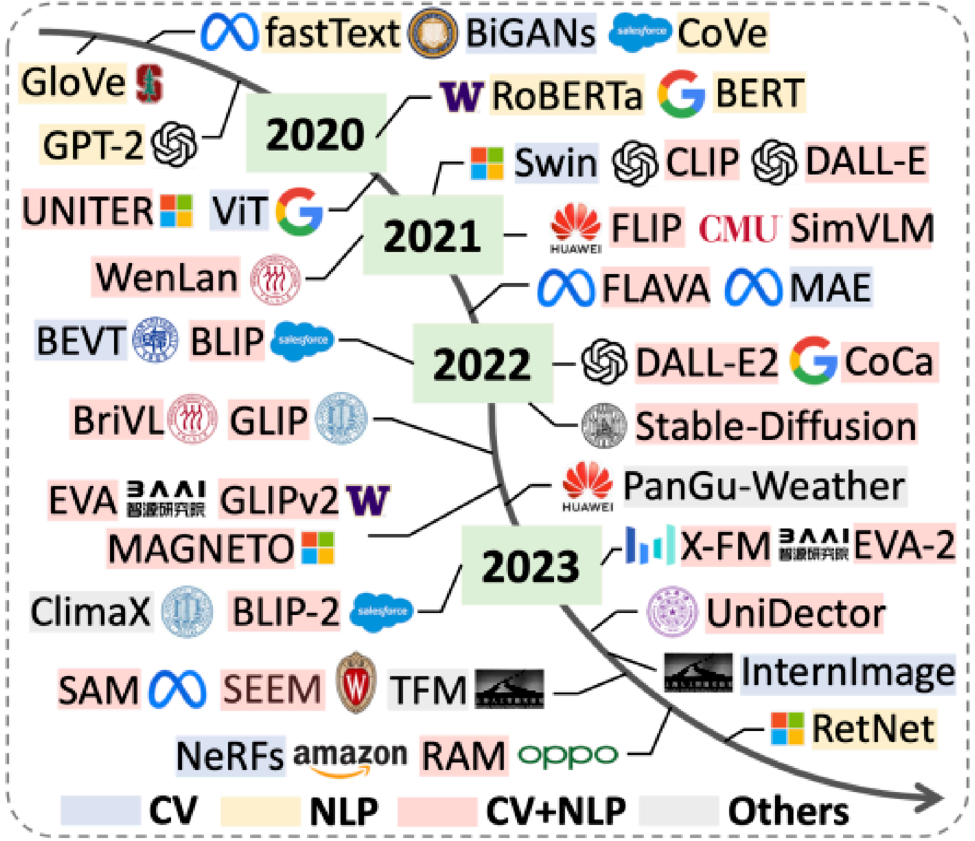
Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., ... & Liang, P. (2021). On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258.

Classification/Categorization

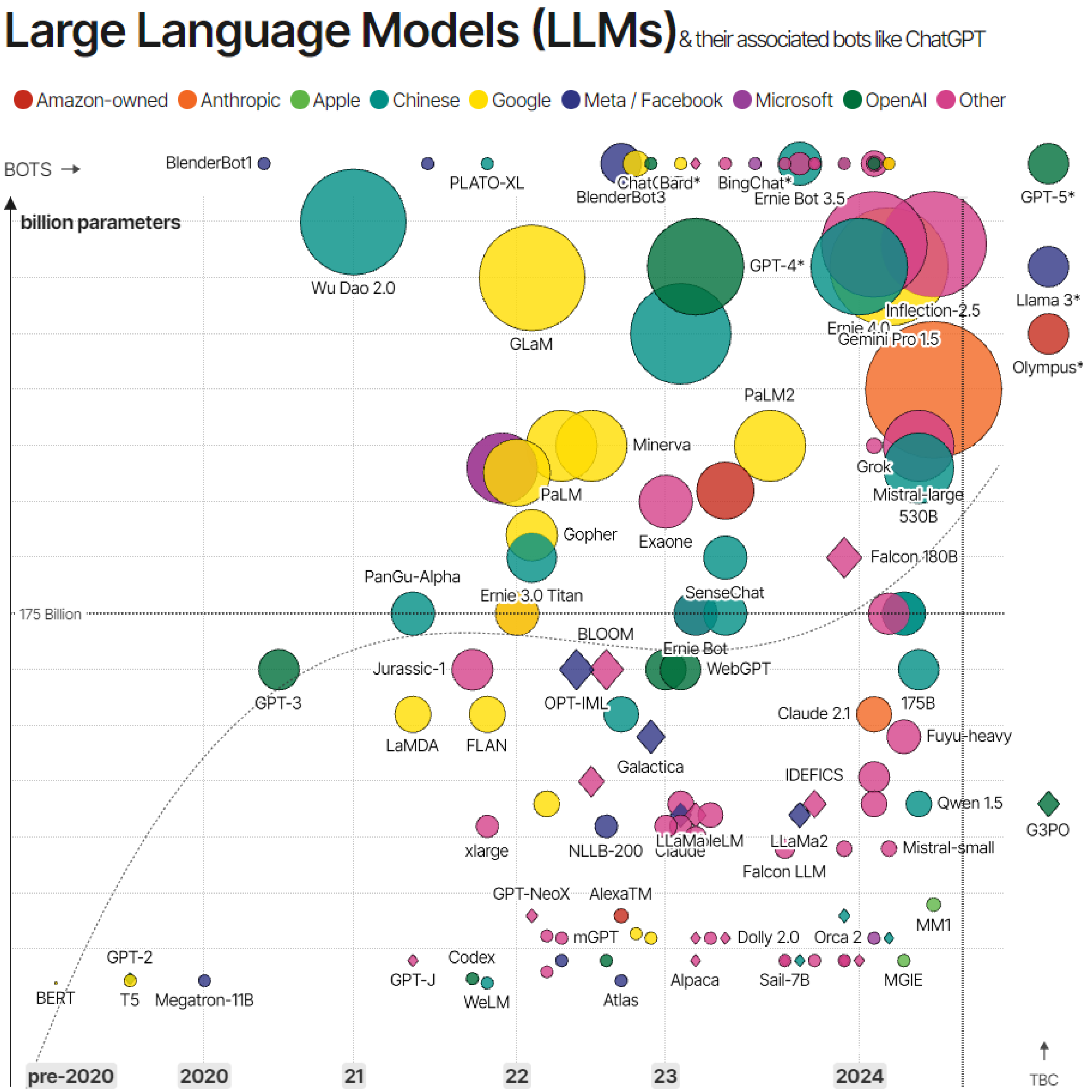
- Large Language Models: Used for natural language processing tasks, such as chatbots, text summarization, and machine translation.
- Vision/Visual Models: Used for computer vision tasks, such as image recognition, segmentation, object detection, and image editing.
- Multimodal Models: Used for tasks involving multiple data types, such as combinations of text, images, videos, and audio.
- Scientific Models: Used for solving scientific problems, such as predicting molecular properties, climate forecasting, and optimizing material design.

What is Foundation Model (5)





Jin, M., Wen, Q., Liang, Y., Zhang, C., Xue, S., Wang, X., ... & Xiong, H. (2023). Large models for time series and spatio-temporal data: A survey and outlook. arXiv preprint arXiv:2310.10196.



[The Rise and Rise of A.I. Large Language Models (LLMs)](#)

Survey

2024

- Image Segmentation in Foundation Model Era: A Survey (from Beijing Institute of Technology)
- Towards Vision-Language Geo-Foundation Model: A Survey (from Nanyang Technological University)
- An Introduction to Vision-Language Modeling (from Meta)
- The Evolution of Multimodal Model Architectures (from Purdue University)
- Efficient Multimodal Large Language Models: A Survey (from Tencent)
- Foundation Models for Video Understanding: A Survey (from Aalborg University)
- Is Sora a World Simulator? A Comprehensive Survey on General World Models and Beyond (from GigaAI)
- Prospective Role of Foundation Models in Advancing Autonomous Vehicles (from Tongji University)
- Parameter-Efficient Fine-Tuning for Large Models: A Comprehensive Survey (from Northeastern University)
- A Review on Background, Technology, Limitations, and Opportunities of Large Vision Models (from Lehigh)
- Large Multimodal Agents: A Survey (from CUHK)
- The Uncanny Valley: A Comprehensive Analysis of Diffusion Models (from Mila)
- Real-World Robot Applications of Foundation Models: A Review (from University of Tokyo)
- From GPT-4 to Gemini and Beyond: Assessing the Landscape of MLLMs on Generalizability, Trustworthiness and Causality through Four Modalities (from Shanghai AI Lab)
- Towards the Unification of Generative and Discriminative Visual Foundation Model: A Survey (from JHU)

Before 2024

- Foundational Models in Medical Imaging: A Comprehensive Survey and Future Vision (from SDSU)
- Multimodal Foundation Models: From Specialists to General-Purpose Assistants (from Microsoft)
- Towards Generalist Foundation Model for Radiology (from SJTU)
- Foundational Models Defining a New Era in Vision: A Survey and Outlook (from MBZ University of AI)
- Towards Generalist Biomedical AI (from Google)
- A Systematic Survey of Prompt Engineering on Vision-Language Foundation Models (from Oxford)
- Large Multimodal Models: Notes on CVPR 2023 Tutorial (from Chunyuan Li, Microsoft)
- A Survey on Multimodal Large Language Models (from USTC and Tencent)
- Vision-Language Models for Vision Tasks: A Survey (from Nanyang Technological University)
- Foundation Models for Generalist Medical Artificial Intelligence (from Stanford)
- A Comprehensive Survey on Pretrained Foundation Models: A History from BERT to ChatGPT
- A Comprehensive Survey of AI-Generated Content (AIGC): A History of Generative AI from GAN to ChatGPT
- Vision-language pre-training: Basics, recent advances, and future trends
- On the Opportunities and Risks of Foundation Models (This survey first popularizes the concept of foundation model; from Standford)

https://github.com/uncbiag/Awesome-Foundation-Models

Main Features of Foundation Models

- Large-Scale Parameters: billions or even tens of billions of parameters. These parameters give the model stronger expressive power

- High Data Demand

- High Computational Complexity

- Improved Performance: Compared to small models, large models often achieve better performance in various tasks. (because large models can learn from more data and capture more complex features and patterns)

- Generalization Ability & Transferability

What makes LLMs/Large Models/Foundation Models popular recently?

- **Increase in Data Volume: With the advancement of the internet and sensor technologies, we can access large-scale data, making it possible to train large models.**

- **Improvement in Computational Power: Due to improvements in hardware and algorithms, it is now possible to effectively train and deploy large-scale deep learning models in distributed systems.**

- **Advancement in Algorithms: Researchers continually improve deep learning algorithms, enabling large models to learn and infer more effectively when handling complex tasks. E.g., Transformers**

How to Get a Foundation Model

Pretraining

- Raw Internet
- Language Modelling
- Base Model
- Thousands of GPUs for months

Supervised Finetuning

- User Demonstrations (assistant responses)
- Language Modelling
- SFT Model
- 1-100 GPUs for days

Reward Modelling

- Comparison Pairs
- Classification
- RM Model
- 1-100 GPUs for days

Reinforcement Learning

- User Instructions (Prompts)
- Reinforcement Learning
- RL Model
- 1-100 GPUs for days

How to Get a Foundation Model: Pretraining

- Massive training data: internet webpages, Wikipedia, books, GitHub, papers, Q&A websites, etc. (low-quality, high-quantity)
- Raw text -> Tokens -> Integers (input ids of the model)
- Language modelling: predict the next token, fill up the masked tokens, denoise
- Thousands of high-performance GPUs, cost in millions
- Large Models: billions of parameters (compress internet-scale into those parameters)
- Base Model, re-trained yearly

Example: GPT-3 (2020)

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or "GPT-3"	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

Table 2.1: Sizes, architectures, and learning hyper-parameters (batch size in tokens and learning rate) of the models which we trained. All models were trained for a total of 300 billion tokens.

- 50,257 vocabulary size
- 2048 context length
- 175B parameters
- Trained on 300B tokens

- Thousands of V100 GPUs
- Months of training
- Millions of USD

Example: LLaMA (2023)

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

Table 1: **Pre-training data.** Data mixtures used for pre-training, for each subset we list the sampling proportion, number of epochs performed on the subset when training on 1.4T tokens, and disk size. The pre-training runs on 1T tokens have the same sampling proportion.

params	dimension	n heads	n layers	learning rate	batch size	n tokens
6.7B	4096	32	32	$3.0e^{-4}$	4M	1.0T
13.0B	5120	40	40	$3.0e^{-4}$	4M	1.0T
32.5B	6656	52	60	$1.5e^{-4}$	4M	1.4T
65.2B	8192	64	80	$1.5e^{-4}$	4M	1.4T

Table 2: **Model sizes, architectures, and optimization hyper-parameters.**

- 32,000 vocabulary size
- 2048 context length
- 65B parameters
- Trained on 1-1.4T tokens

- 2,048 A100 GPUs
- 21 days of training
- $5M USD

How to Get a Foundation Model: Supervised Finetuning

- Initialized from Base Model
- Instructions, demonstrations, ideal assistant responses (prompt + responses, low-quantity, high-quality)
- Same algorithms, e.g., predict the next token
- Ability to understand instructions
- Less GPUs, high impact
- SFT Model, re-trained weekly/monthly

How to Get a Foundation Model: Reward Modelling

- Initialized from SFT Model
- 100k-1M comparisons from crowdsourcing annotators (low-quantity, high-quality)
- Goal: to construct a text quality comparison model, which ranks the quality of multiple different output results given by the SFT model for the same prompt.
- RM can judge the superiority or inferiority between two input results through a classification model, e.g., good output vs. bad output
- RM model itself cannot be used independently by users
- Challenge: maintaining consistency among crowdsourced annotators

How to Get a Foundation Model: Reinforcement Learning

- Initialized from SFT Model use RM
- 10k-100k prompts from crowdsourcing annotators (low-quantity, high-quality)
- Aligning language models with user intent, mitigating the harms of language models
- Optimize a policy against the reward model, fine-tune the supervised policy to optimize the reward (maximize the reward score of policy generated output)
- Challenge: reduce the “creativity”

Alignment

- **May yield texts that are imprecise, misleading, or even harmful.**
- **So, need to employ alignment techniques to ensure these models to exhibit behaviours consistent with human values**

- **What is Alignment?**
- **IBM: Alignment is the process of encoding human values and goals into large language models to make them as helpful, safe, and reliable as possible.**

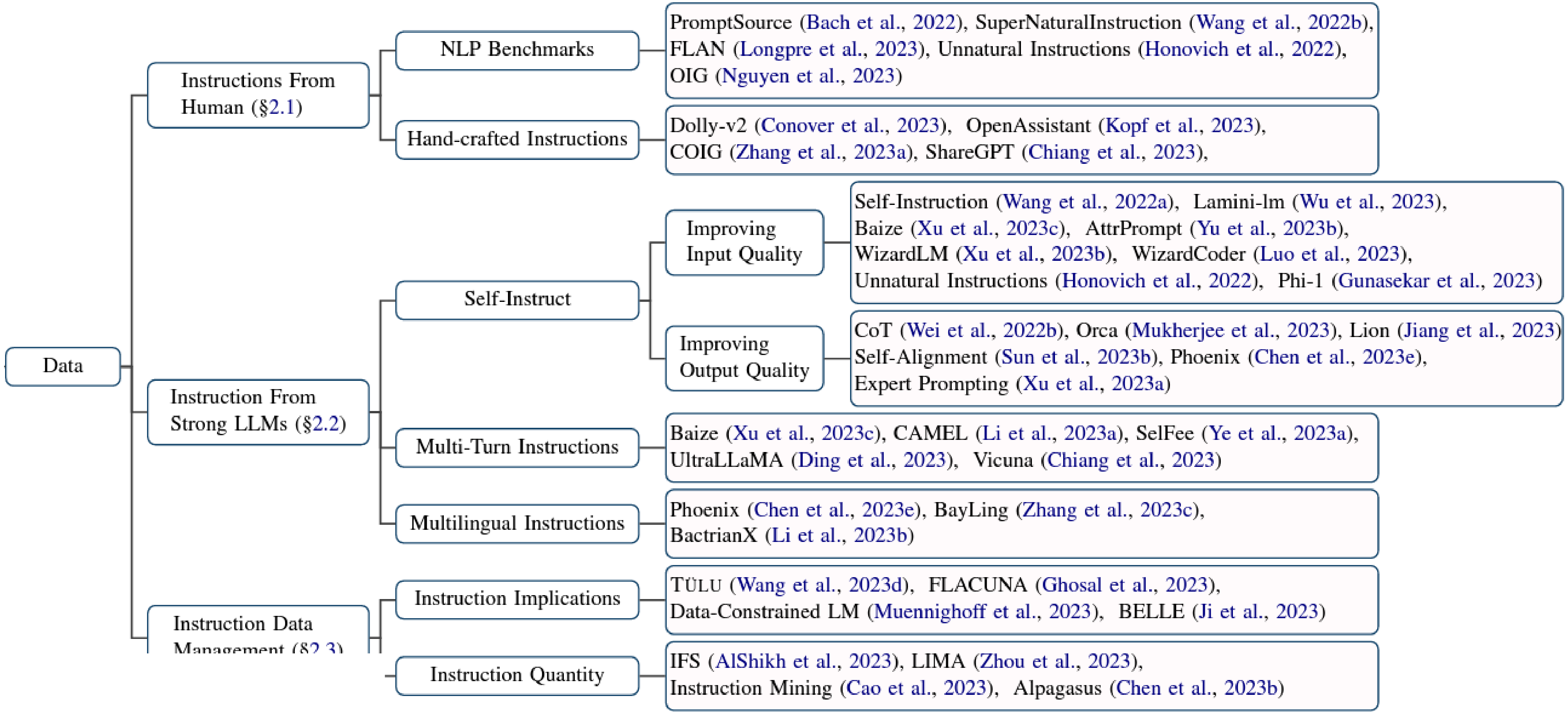
Alignment

- Collecting high quality data for both SFT and RLHF stages can be costly and time-consuming.

- The training strategies need to be optimized as SFT training is resource-consuming, and reinforcement learning in RLHF often lacks stability.

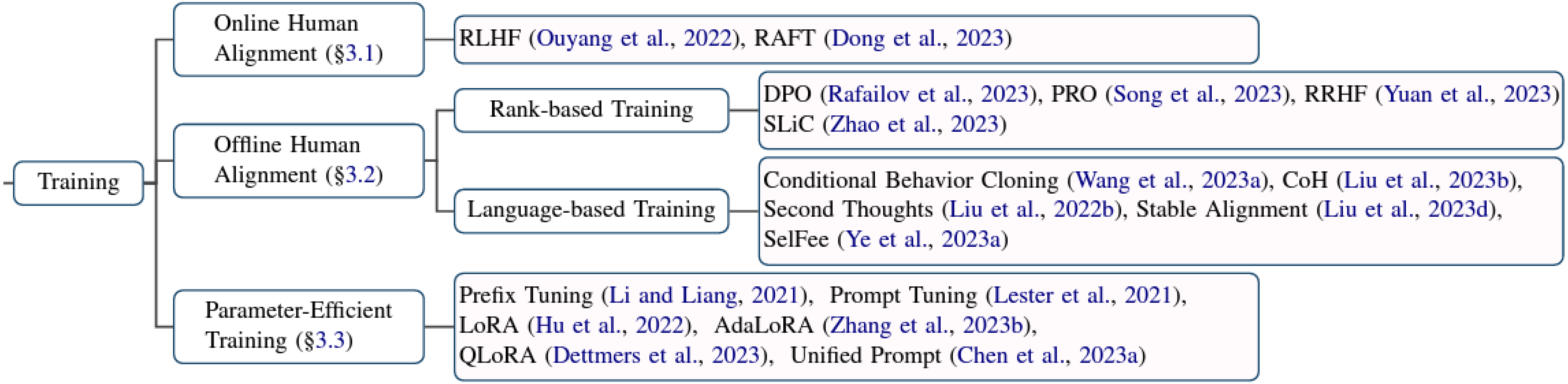
- Evaluating LLMs comprehensively is challenging

Alignment: Data



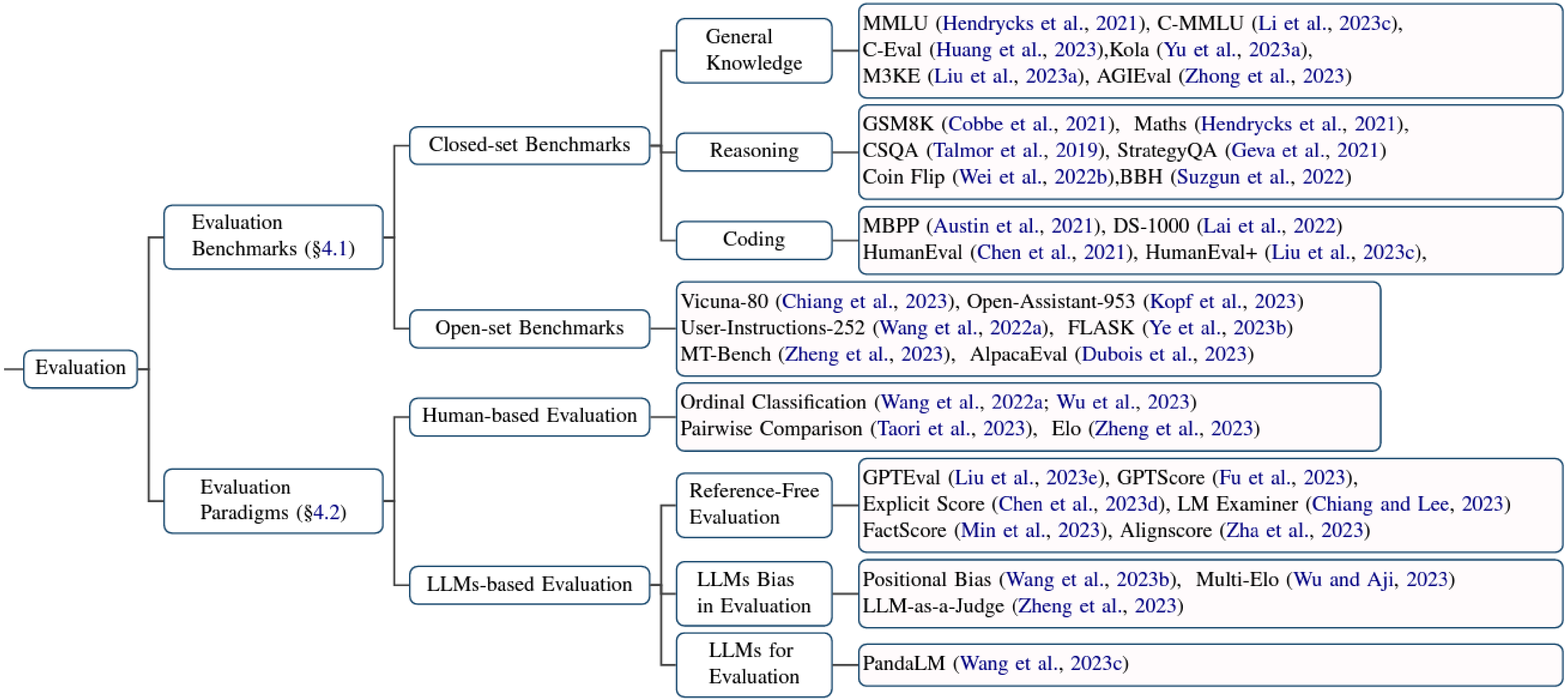
Wang, Y., Zhong, W., Li, L., Mi, F., Zeng, X., Huang, W., ... & Liu, Q. (2023). Aligning large language models with human: A survey. *arXiv preprint arXiv:2307.12966*.

Alignment: Training



Wang, Y., Zhong, W., Li, L., Mi, F., Zeng, X., Huang, W., ... & Liu, Q. (2023). Aligning large language models with human: A survey. *arXiv preprint arXiv:2307.12966*.

Alignment: Evaluation



Wang, Y., Zhong, W., Li, L., Mi, F., Zeng, X., Huang, W., ... & Liu, Q. (2023). Aligning large language models with human: A survey. *arXiv preprint arXiv:2307.12966*.

Alignment: RLHF

- Reinforcement Learning from Human Feedback (RLHF) is a method used to align AI systems, including LLMs, by incorporating human feedback into their training process.

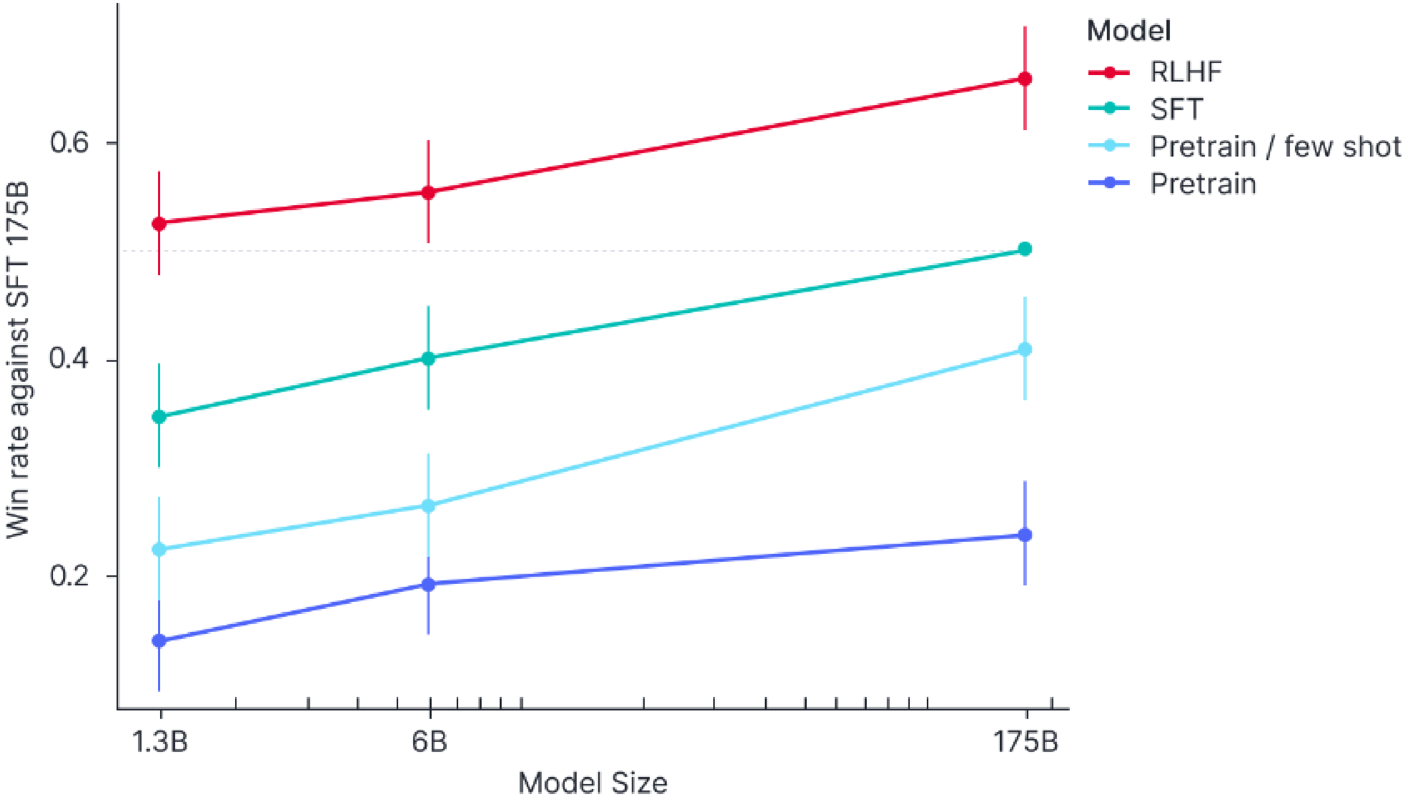
- RLHF leverages human preferences to fine-tune the outputs.

- Humans provide feedback on the model's behaviour, and this feedback is used to improve the model's performance and alignment with desired outcomes.

- Ensures that AI outputs align more closely with human values and intentions

- Reduces the risk of harmful or biased behaviour

How to Get a Foundation Model: Reinforcement Learning



Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., ... & Lowe, R. (2022). Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, *35*, 27730-27744.

How to Get a Foundation Model

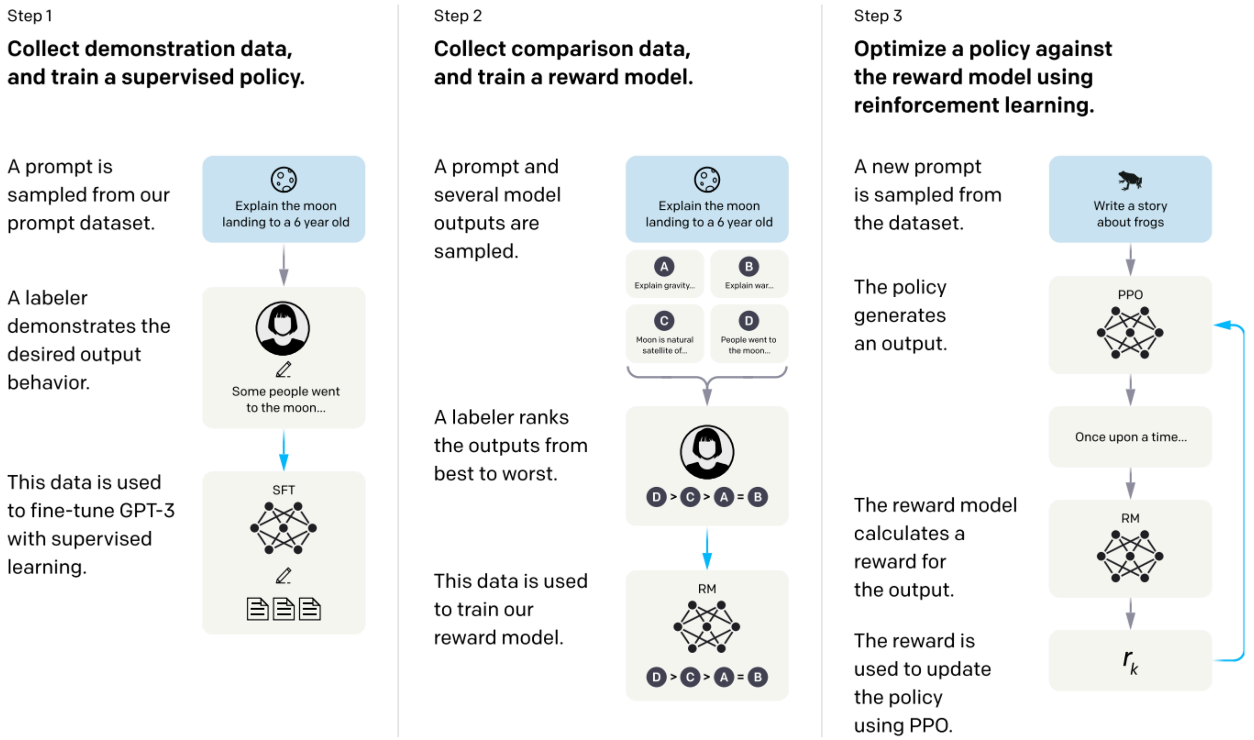


Figure 2: A diagram illustrating the three steps of our method: (1) supervised fine-tuning (SFT), (2) reward model (RM) training, and (3) reinforcement learning via proximal policy optimization (PPO) on this reward model. Blue arrows indicate that this data is used to train one of our models. In Step 2, boxes A-D are samples from our models that get ranked by labelers.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., ... & Lowe, R. (2022). Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, *35*, 27730-27744.

Alignment: RLHF Limitations

- **Human Effort: not easily scalable**
- **Inconsistent Judgments: Human evaluators may have subjective opinions**
- **Human Bias Transfer**
- **Risk of Overfitting to Feedback**

How to Get a Foundation Model

Foundation Model Development Cheatsheet

Resources and recommendations for best practices in developing and releasing models.

This cheatsheet serves as a succinct guide, prepared *by* foundation model developers *for* foundation model developers. As the field of AI foundation model development rapidly expands, welcoming new contributors, scientists, and applications, we hope to lower the barrier for new community members to become familiar with the variety of resources, tools, and findings. The focus of this cheatsheet is not only, or even primarily, to support building, but to inculcate good practices, awareness of limitations, and general responsible habits as community norms.

To add to the cheatsheet please see Contribute Resources.

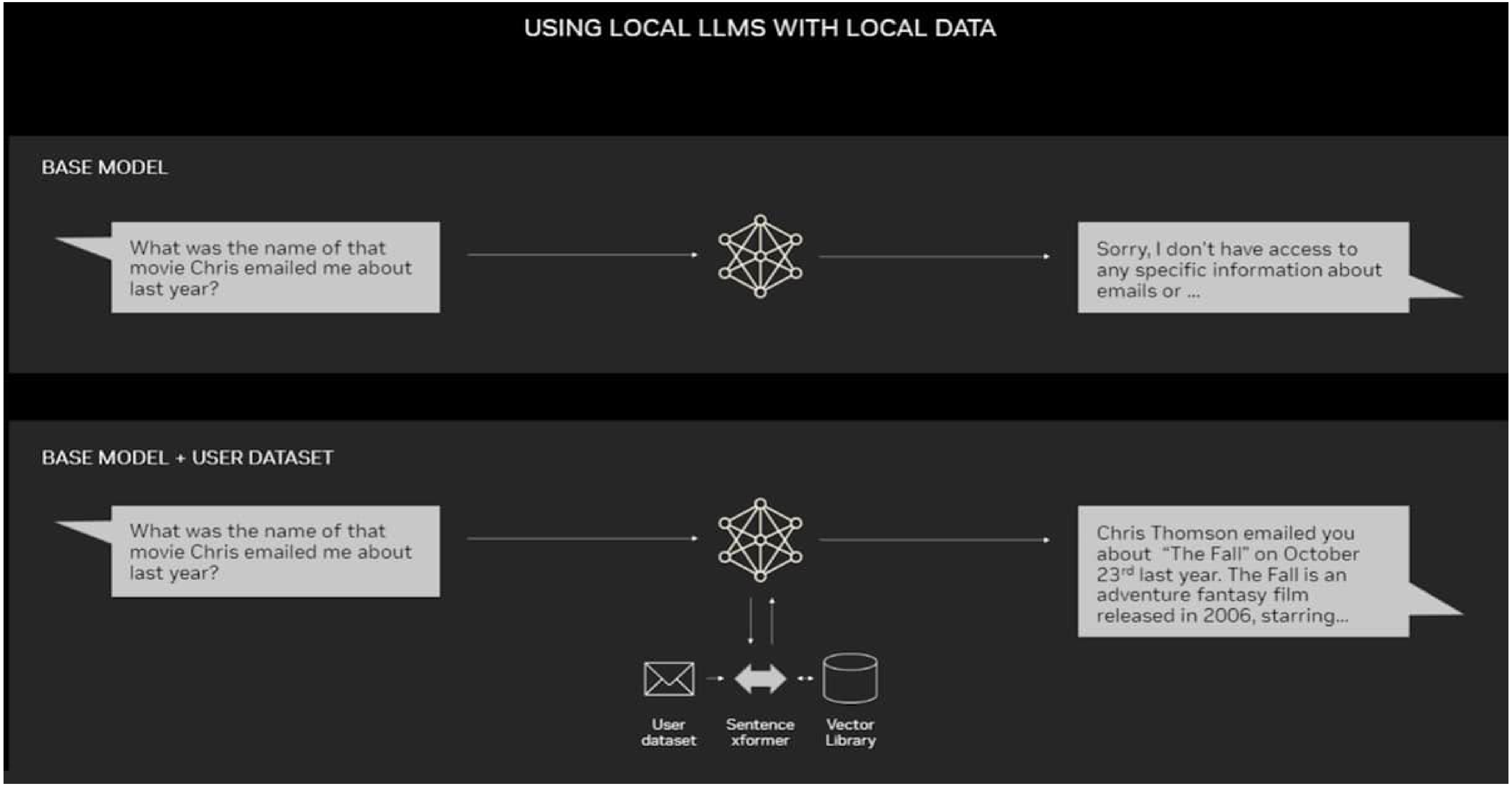
https://fmcheatsheet.org/

https://www.borealisai.com/research-blogs/training-and-fine-tuning-large-language-models/

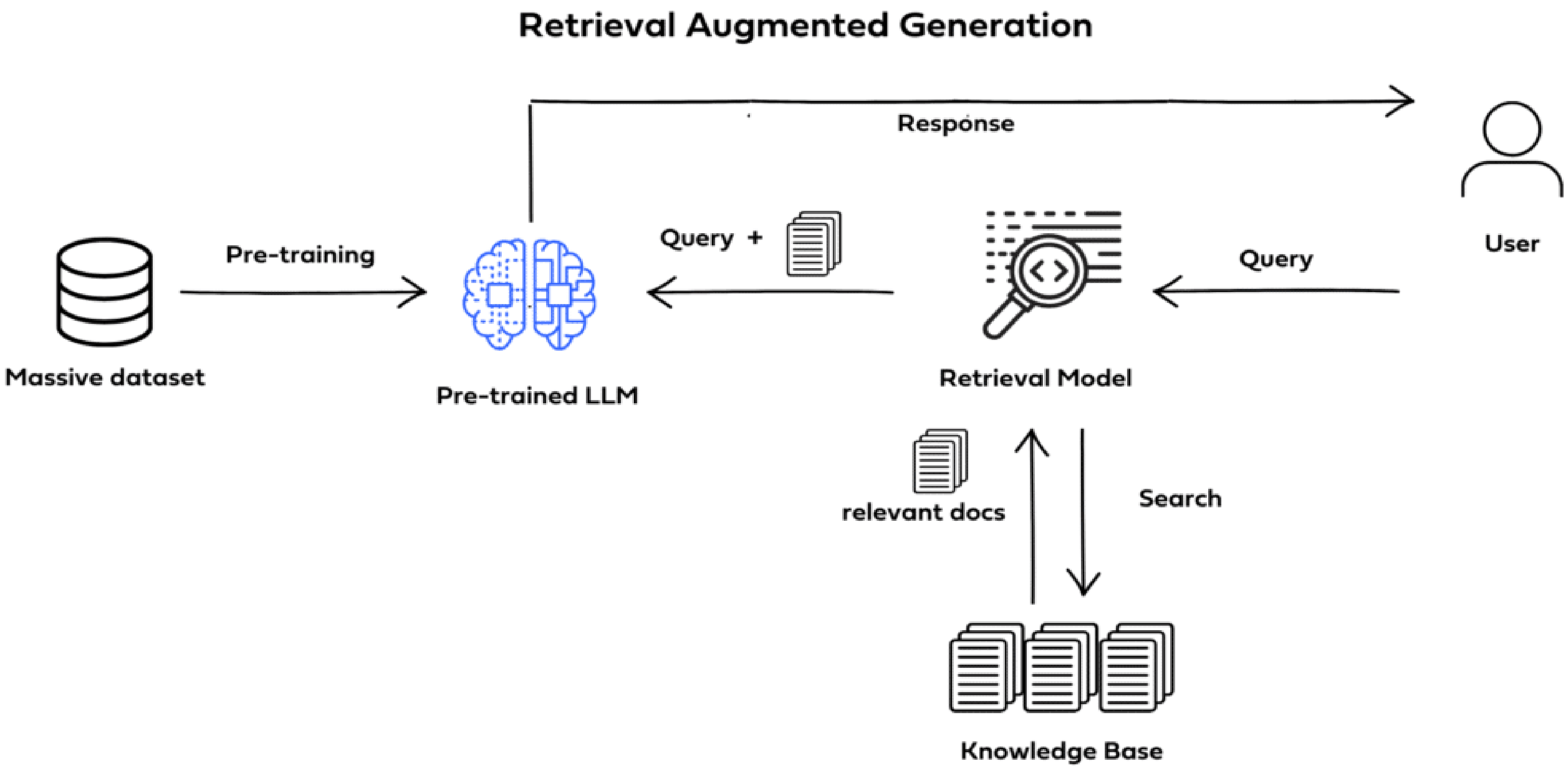
Challenges of LLMs

- Presenting false information when it does not have the answer.
- Presenting out-of-date or generic information when the user expects a specific, current response.
- Creating a response from non-authoritative sources.
- Creating inaccurate responses due to terminology confusion, wherein different training sources use the same terminology to talk about different things.
- Less control over the content
- Data Privacy and Security

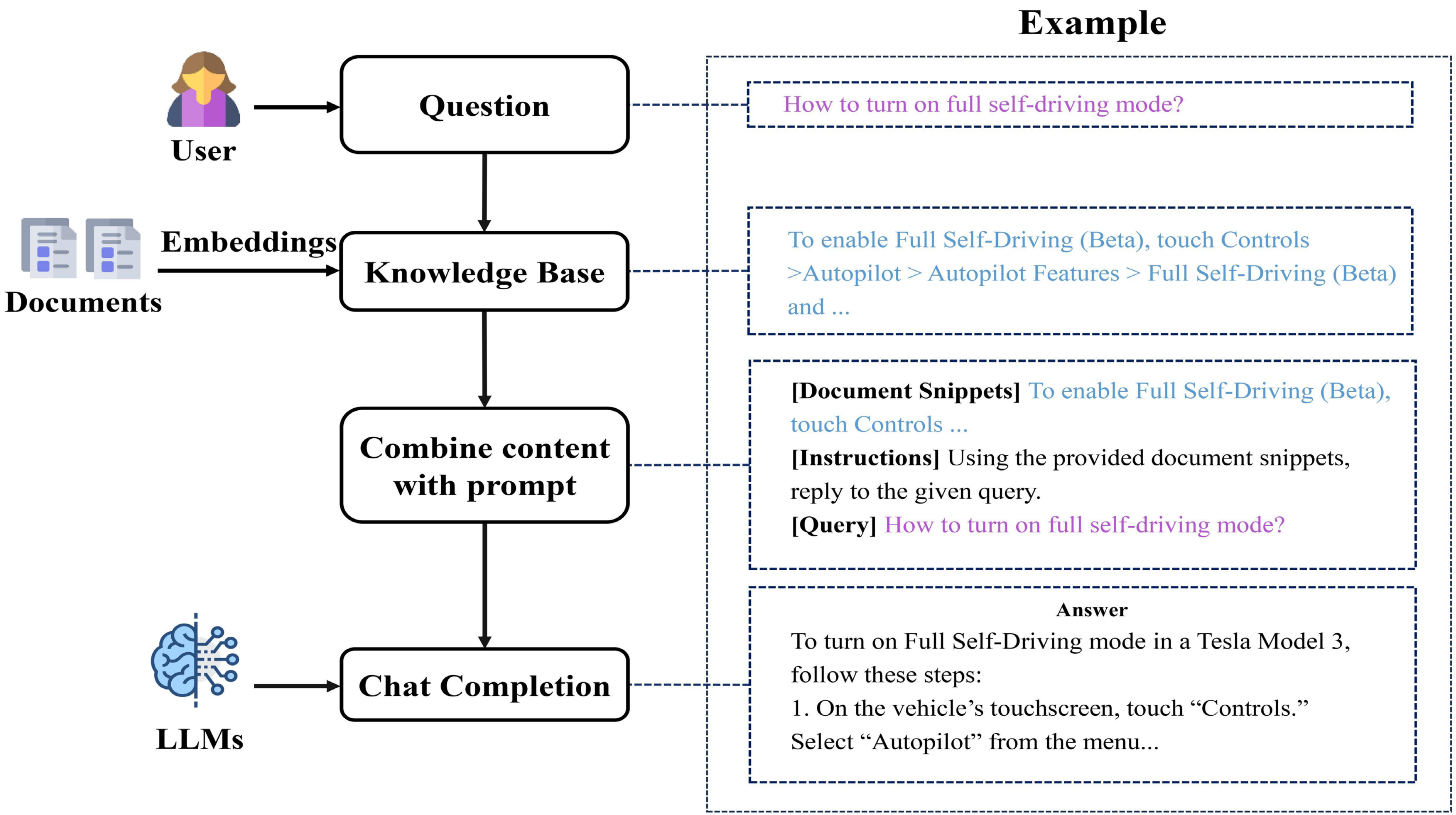
Challenges of LLMs



Retrieval Augmented Generation (RAG)



Retrieval Augmented Generation (RAG)



Some criticism

- Are bigger language models better?
  - Bigger models are better at generalizing on various downstream tasks
  - Issues: Bias, carbon footprints

ARTICLE OPEN ACCESS

On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? 🦜

Authors: Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, Shmargaret Shmitchell Authors Info & Claims

FAccT '21: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency • March 2021 • Pages 610–623 • https://doi.org/10.1145/3442188.3445922

Published: 03 March 2021

Large computer language models carry environmental, social risks

It takes an enormous amount of computing power to fuel the model language programs, Bender said. That takes up energy at tremendous scale, and that, the authors argue, causes environmental degradation. And those costs aren't borne by the computer engineers, but rather by marginalized people who cannot afford the environmental costs.

Another risk comes from the training data itself, the authors say. Because the computers read language from the Web and from other sources, they can pick up and perpetuate racist, sexist, ableist, extremist and other harmful ideologies.

Yann LeCun
27 October 2020 at 13:01 ·

"It's entertaining, and perhaps mildly useful as a creative help," LeCun wrote. "But trying to build intelligent machines by scaling up language models is like building high-altitude airplanes to go to the moon. You might beat altitude records, but going to the moon will require a completely different approach."

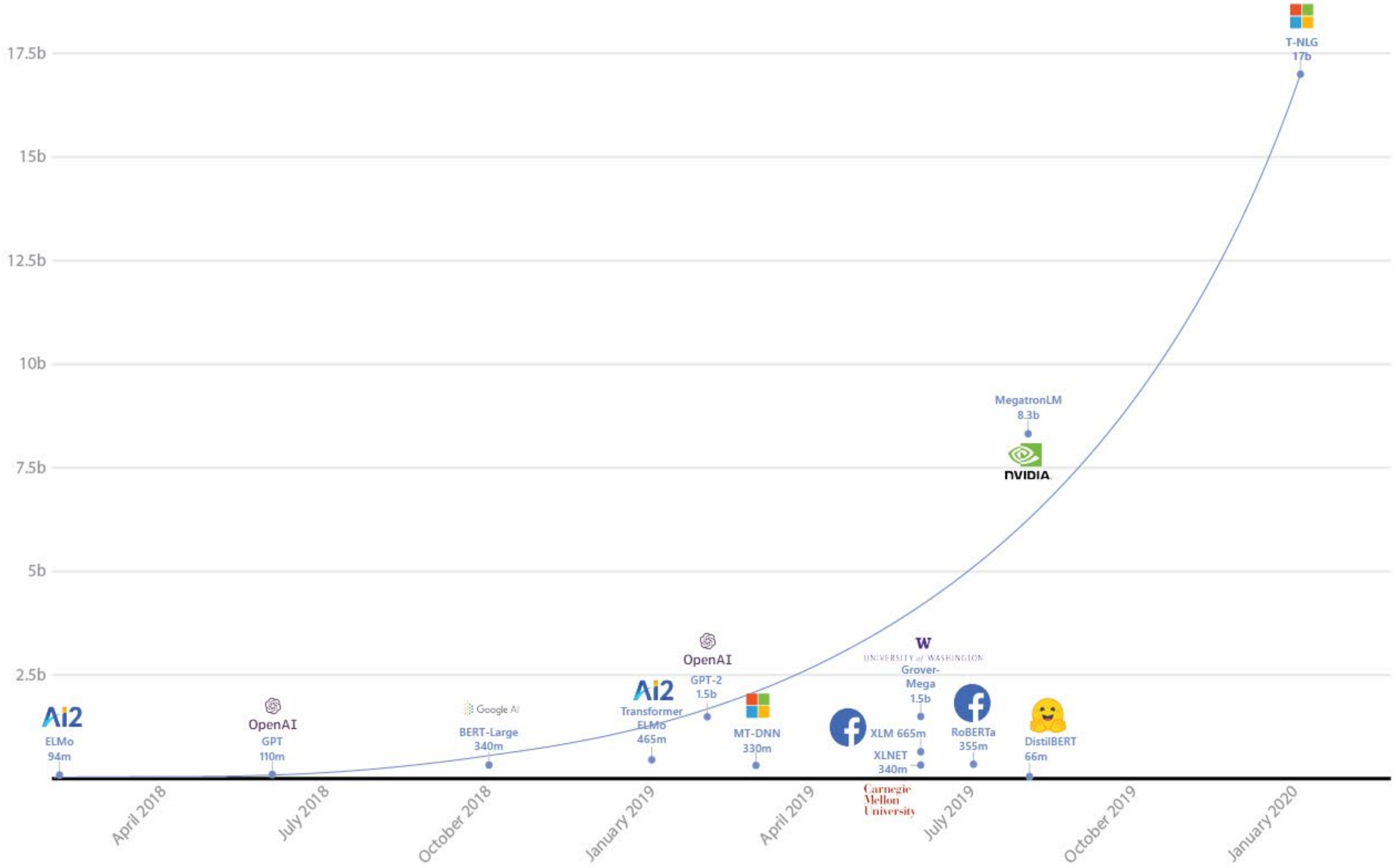
Criticism

Deep Learning, Deep Pockets?

As you would expect, training a 530-billion parameter model on humongous text datasets requires a fair bit of infrastructure. In fact, Microsoft and NVIDIA used hundreds of DGX A100 multi-GPU servers. At $199,000 a piece, and factoring in networking equipment, hosting costs, etc., anyone looking to replicate this experiment would have to spend close to $100 million dollars. Want fries with that?

Turing-NLG

- A Transformer based generative language model
- Based on DeepSpeed (DL library) to make distributed training of large models easier
- 17 billion parameters



OpenAI GPT-3

- Key idea: Improve task-agnostic few-shot performance
- Evaluation on various NLP tasks under few-shot learning, one-shot learning, and zero-shot learning demonstrates GPT-3 promising results
- Technical details:
  - An autoregressive language model
  - GPT-3 has 96 layers with each layer having 96 attention heads
  - trained on datasets with 500 billion tokens
  - Word embedding size of 12888
  - Context window size is of 2048 tokens
  - Uses alternating dense and locally banded sparse attention patterns
- Compute:
  - Trained on more than 576 GB of text data including common crawl, Books, and Wikipedia
  - About 175 billion parameters
  - costs OpenAI around $4.6 million

Credit: Brown et al., (2020). Language models are few-shot learners.
Priya Shree. Medium: The journey of OpenAI GPT models. https://medium.com/walmartglobaltech/the-journey-of-open-ai-gpt-models-32d95b7b7fb2

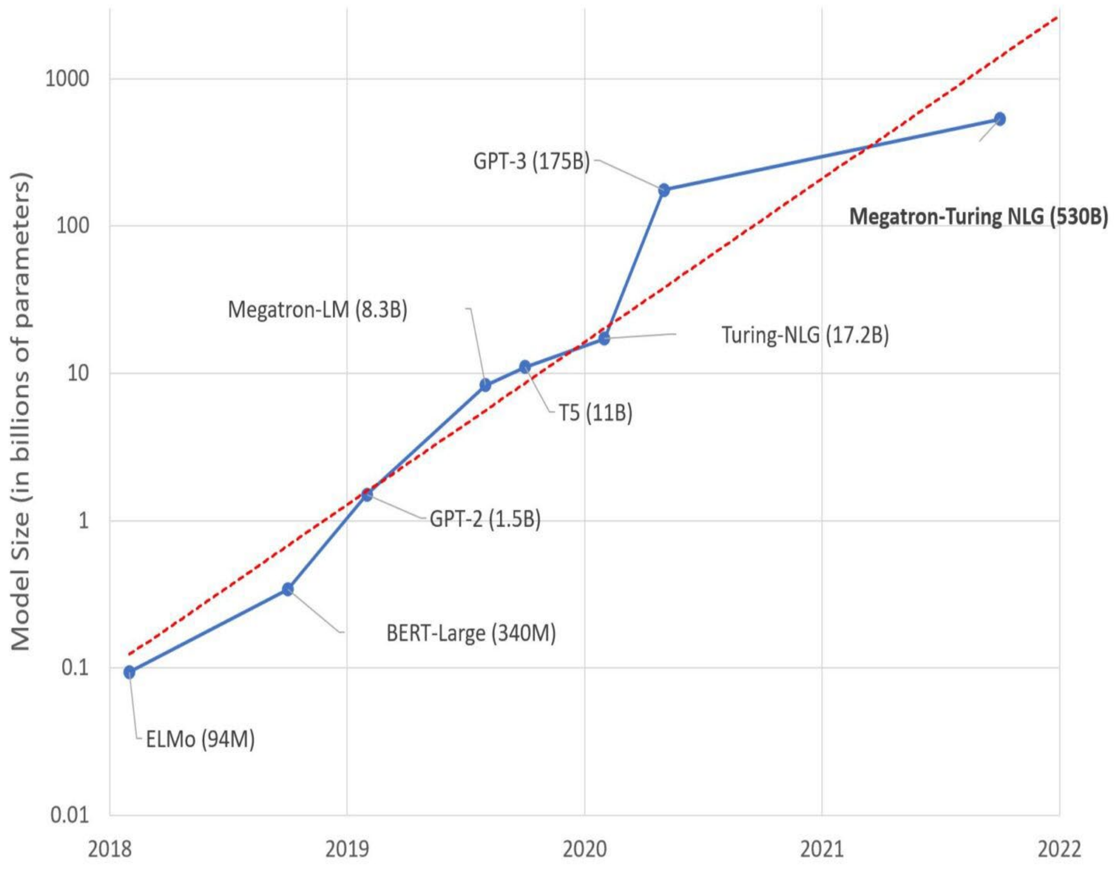
MT-NLG

Behold the Megatron: Microsoft and Nvidia build massive language processor

MT-NLG is a beast that fed on over 4,000 GPUs

Katyanna Quach   Tue 12 Oct 2021 // 00:36 UTC

- Largest transformer model to date
- 530 billion parameters
- Trained on a wide variety of NLP tasks such as auto-completing sentences, question and answering, reading and reasoning
- Outperforms various tasks with little to no fine-tuning, in a few-shot or zero-shot learning
- Trained using Nvidia's Selene machine learning supercomputer
- Estimated to cost $85 million



OpenAI GPT-4o

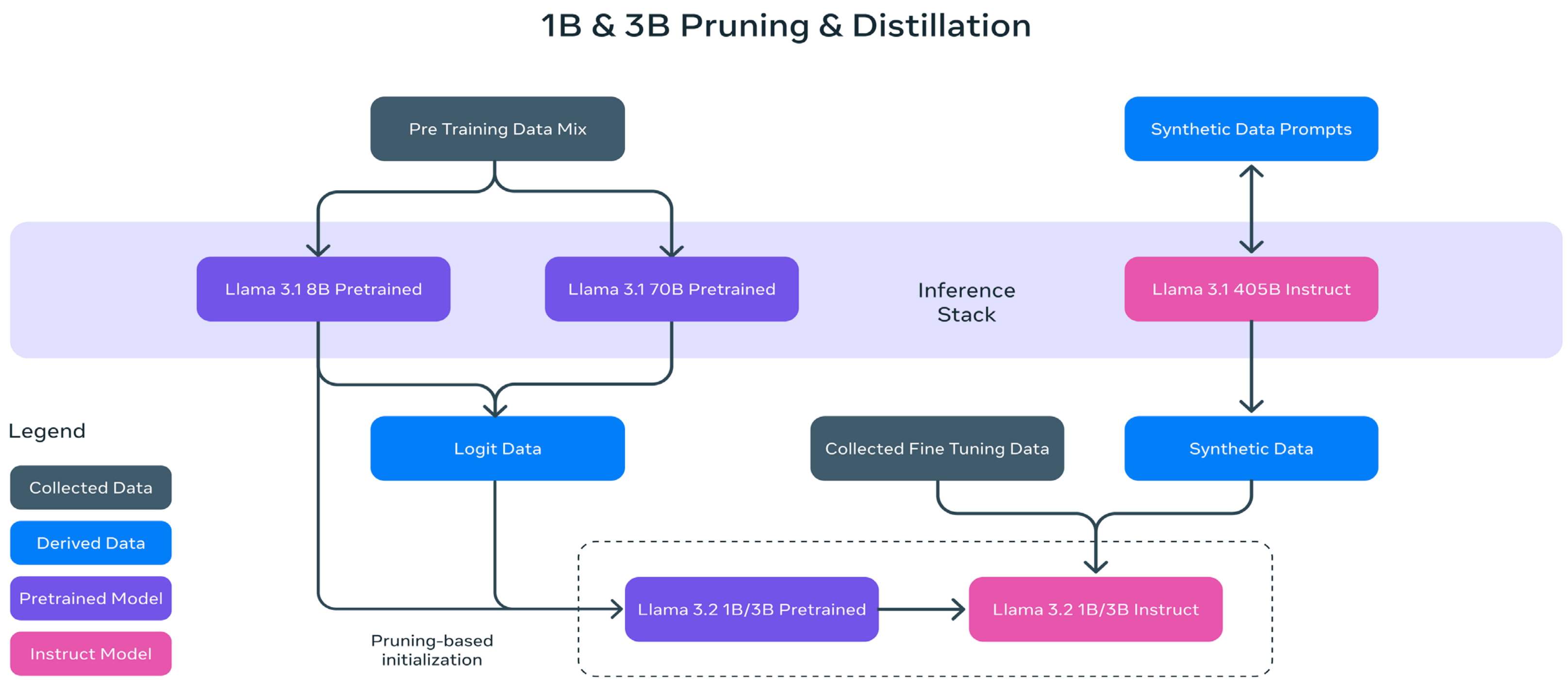
- GPT-4o (Omni or Universal) https://openai.com/index/hello-gpt-4o/
- Multimodal, Multi-lingual Generative Pre-trained Transformer Model
- Released in May 2024

Llama 3.2 https://www.llama.com

- Meta's open-source AI model you can fine-tune, distill, and deploy.
- Llama 3.2 is a collection of **large language models (LLMs) pretrained and fine-tuned in 1B and 3B sizes that are multilingual text only**, and **11B and 90B sizes that take both text and image inputs and output text**.
- Evaluated performance on over 150 benchmark datasets that span a wide range of languages. For the vision LLMs, evaluated performance on benchmarks for image understanding and visual reasoning.
- Released September, 2024
- Llama 3.2 1B and 90B models support image reasoning use cases, such as document-level understanding including charts and graphs, captioning of images, and visual grounding tasks such as directionally pinpointing objects in images based on natural language descriptions.

Llama 3.2 https://www.llama.com

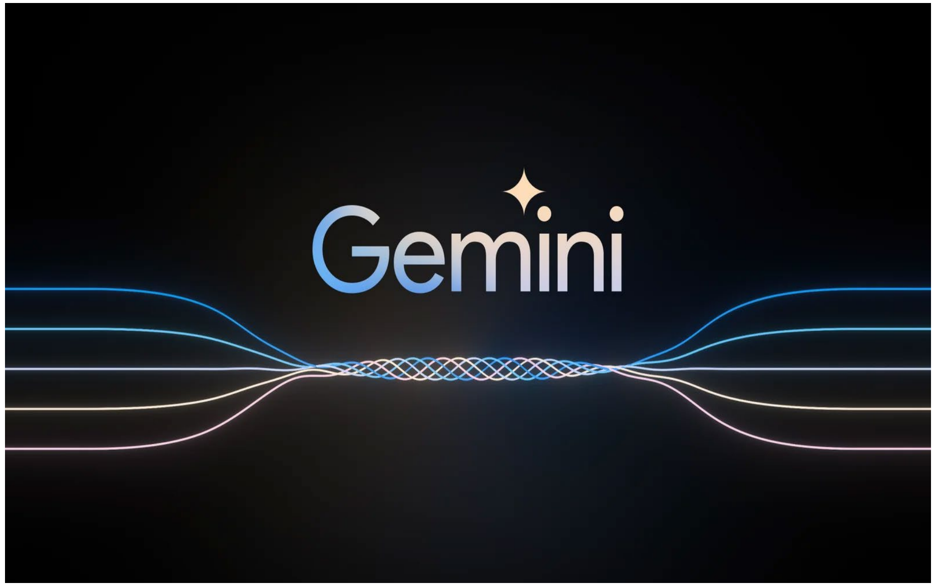
- Key difference between the predecessors' models:
  - size of the pretraining corpus increased by 650% (LLaMA2 was trained on 2T tokens where as LLaMA3 trained on 15T tokens)
  - doubled the context length of the model from 4K to 8K on both 8B and 70B models,
  - adopted grouped-query attention (GQA) for both 8B and 70B variant as compared to the previous generations.



Gemini https://gemini.google.com/?hl=en-AU

- Google's largest and most capable AI model
- Natively multimodal - it can generalize and seamlessly understand, operate across and combine different types of information including text, code, audio, image and video.

- With a score of 90.0%, Gemini Ultra is the first model to outperform human experts on MMLU(massive multitask language understanding), which uses a combination of 57 subjects such as math, physics, history, law, medicine and ethics for testing both world knowledge and problem-solving abilities.



Gemini’s performance

TEXT

Capability	Benchmark Higher is better	Description	Gemini Ultra	GPT-4 API numbers calculated where reported numbers were missing
General	MMLU	Representation of questions in 57 subjects (incl. STEM, humanities, and others)	90.0% CoT@32*	86.4% 5-shot** (reported)
Reasoning	Big-Bench Hard	Diverse set of challenging tasks requiring multi-step reasoning	83.6% 3-shot	83.1% 3-shot (API)
	DROP	Reading comprehension (F1 Score)	82.4 Variable shots	80.9 3-shot (reported)
	HellaSwag	Commonsense reasoning for everyday tasks	87.8% 10-shot*	95.3% 10-shot* (reported)
Math	GSM8K	Basic arithmetic manipulations (incl. Grade School math problems)	94.4% maj1@32	92.0% 5-shot CoT (reported)
	MATH	Challenging math problems (incl. algebra, geometry, pre-calculus, and others)	53.2% 4-shot	52.9% 4-shot (API)
Code	HumanEval	Python code generation	74.4% 0-shot (IT)*	67.0% 0-shot* (reported)
	Natural2Code	Python code generation. New held out dataset HumanEval-like, not leaked on the web	74.9% 0-shot	73.9% 0-shot (API)

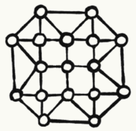
* See the technical report for details on performance with other methodologies
** GPT-4 scores 87.29% with CoT@32 - see the technical report for full comparison

MULTIMODAL

Capability	Benchmark	Description Higher is better unless otherwise noted	Gemini	GPT-4V Previous SOTA model listed when capability is not supported in GPT-4V
Image	MMMU	Multi-discipline college-level reasoning problems	59.4% 0-shot pass@1 Gemini Ultra (pixel only*)	56.8% 0-shot pass@1 GPT-4V
	VQAv2	Natural image understanding	77.8% 0-shot Gemini Ultra (pixel only*)	77.2% 0-shot GPT-4V
	TextVQA	OCR on natural images	82.3% 0-shot Gemini Ultra (pixel only*)	78.0% 0-shot GPT-4V
	DocVQA	Document understanding	90.9% 0-shot Gemini Ultra (pixel only*)	88.4% 0-shot GPT-4V (pixel only)
	Infographic VQA	Infographic understanding	80.3% 0-shot Gemini Ultra (pixel only*)	75.1% 0-shot GPT-4V (pixel only)
	MathVista	Mathematical reasoning in visual contexts	53.0% 0-shot Gemini Ultra (pixel only*)	49.9% 0-shot GPT-4V
Video	VATEX	English video captioning (CIDEr)	62.7 4-shot Gemini Ultra	56.0 4-shot DeepMind Flamingo
	Perception Test MCQA	Video question answering	54.7% 0-shot Gemini Ultra	46.3% 0-shot SeViLA
Audio	CoVoST 2 (21 languages)	Automatic speech translation (BLEU score)	40.1 Gemini Pro	29.1 Whisper v2
	FLEURS (62 languages)	Automatic speech recognition (based on word error rate, lower is better)	7.6% Gemini Pro	17.6% Whisper v3

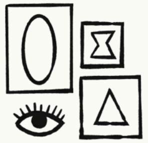
Anthropic Claude 3.5 https://www.anthropic.com/claude

Claude's capabilities



Advanced reasoning

Claude can perform complex cognitive tasks that go beyond simple pattern recognition or text generation

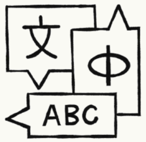


Vision analysis

Transcribe and analyze almost any static image, from handwritten notes and graphs, to photographs

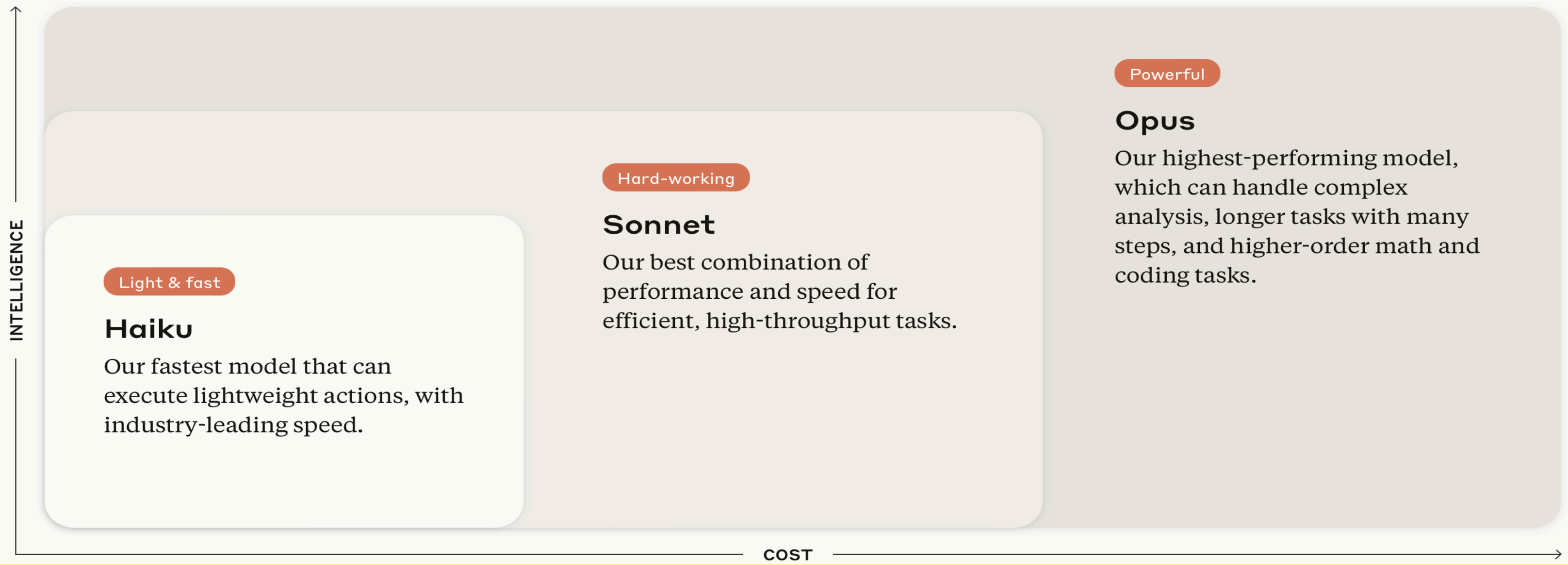
Code generation

Start creating websites in HTML and CSS, turning images into structured JSON data, or debugging complex code bases



Multilingual processing

Translate between various languages in real-time, practice grammar, or create multi-lingual content



Mistral AI https://mistral.ai/technology/#models

- Build with open-weight models

Mistral Small 24.09

Enterprise-grade small model.

- The most powerful model in its size
- Available under the Mistral Research License
- 128k token context window
- Cost-efficient and fast model for a wide array of use cases such as translation, summarization, and sentiment analysis

Mistral Large 2

Top-tier reasoning for high-complexity tasks, for your most sophisticated needs.

- Multi-lingual (incl. European languages, Chinese, Japanese, Korean, Hindi, Arabic)
- Large context window of 128K tokens
- Native function calling capacities and JSON outputs
- High coding proficiency (80+ coding languages)

Codestral

State-of-the-art Mistral model trained specifically for code tasks.

- Trained on 80+ programming languages (incl. Python, Java, C, C++, PHP, Bash)
- Optimized for low latency: Way smaller than competitive coding models
- Context window of 32K tokens

Mistral Embed

State-of-the-art semantic for extracting representation of text extracts.

- English only for now
- Achieves a retrieval score of 55.26 on the Massive Text Embedding Benchmark (MTEB)

Ministral 3B 24.10

Our most efficient edge model.

The most capable in its category, ideal for low-power, low-latency on-device computing and edge use cases.

- 128k token context window
- Highly capable in function-calling for agentic workflows
- Ideal for lossless quantization and tailoring specific use cases

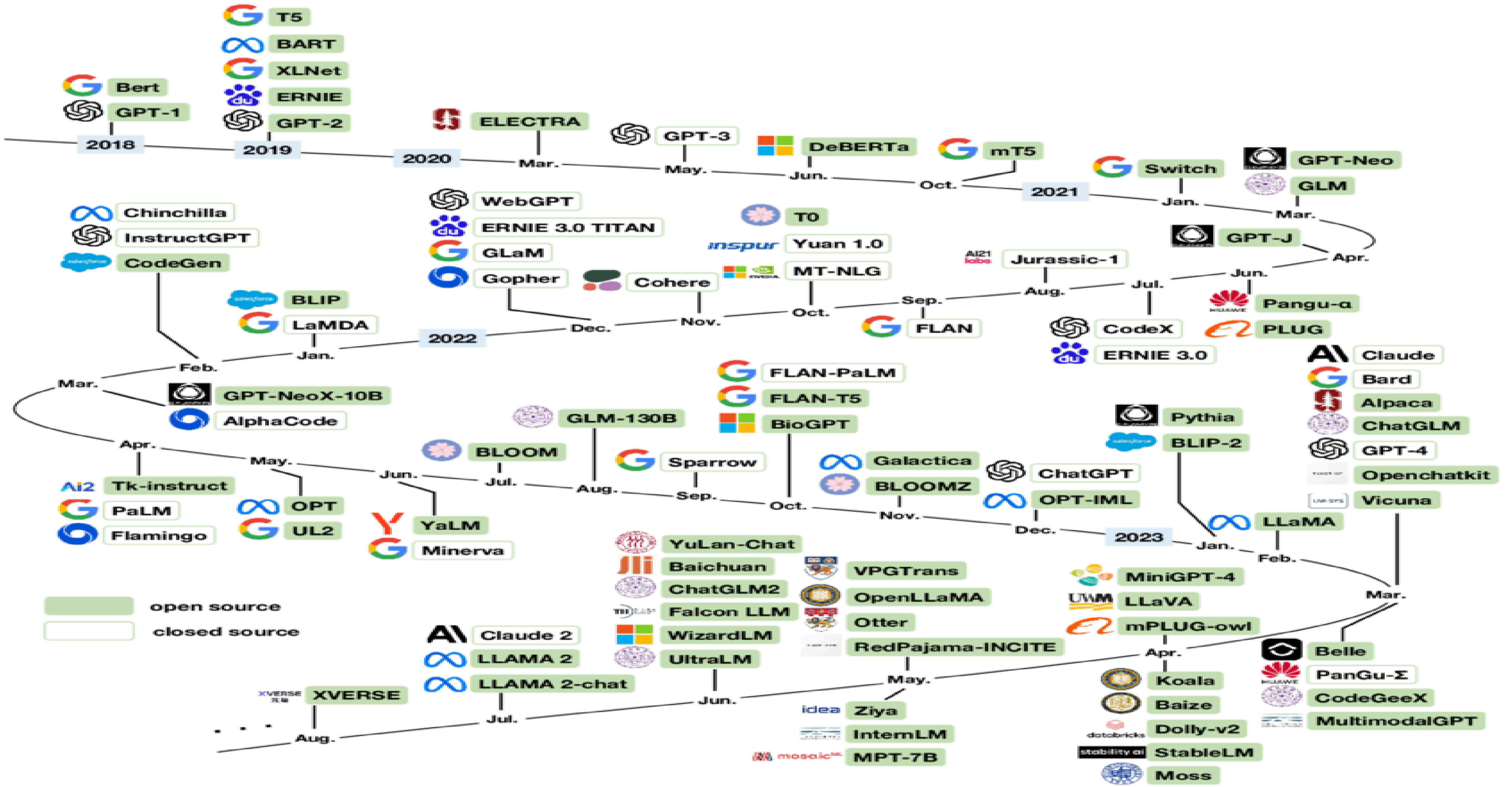
Ministral 8B 24.10

Our most powerful edge model. Successor to Mistral 7B.

Sets the benchmark in commonsense, reasoning, and function-calling in the sub-10B category.

- 128k token context window
- Ideal for on-device computing and edge use cases
- Highly capable in reasoning, function-calling, and common sense

LLMs over the last few years …



Summary

- Large language modeling is like race between titans
- Large language model size is increasing 10x every year for the last few years. Moore's Law
- Issues:
  - Environmental cost (carbon footprint)
  - Hard to replicate
  - Very complex
- The need:
  - To build practical and efficient solutions
  - Within everyone's capability and reach to solve real-world problems

Credit: Brown et al., (2020). Language models are few-shot learners.
Priya Shree. Medium: The journey of OpenAI GPT models. https://medium.com/walmartglobaltech/the-journey-of-open-ai-gpt-models-32d95b7b7fb2

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Questions?